READ ME FIRST —
Summary of the Research So Far

Learned Topology in Next-Token Transformers

Do small transformers trained on hidden-walk sequences represent, explain and use the topology of the latent world? (project topo-pilot)

The questions, the methods, every experiment to date (the pilot, the studies of where the learned circle comes from and whether the model uses it, the prediction-horizon studies, and the crossing-order studies), the results including the negative and inconclusive ones, and what remains open

21 October 2026

Living document; updated as runs complete.

Compiled from the repository (`topo-pilot`, branch `main`), the coordination mailbox (`agent-handoff`) and the execution stores beside it. Every number is taken from a cited file. The crossing-order study on trained models reached its preregistered decision on 21 October 2026; its decision-stage report is cited where its numbers are used. This text changes no decision. Where it summarises, the cited reviews, reports and frozen protocols are authoritative.

Abstract

I ask whether a small causal transformer, trained from scratch only to predict the next symbol of sequences emitted by a hidden random walk on a world of known topology, develops internal structure that reflects that topology, where such structure comes from, and whether the model uses it. The pilot compares a 12×12 cylinder with a 12×12 square under matched dynamics and emissions, with an exact Bayesian oracle, matched emission controls, and representation baselines, and decides one primary hypothesis by a rule frozen before any model was trained. Under the smooth emission, training produces a persistent-H_1 signal in the final residual stream that separates the cylinder from the square in 24 of 24 panels across three seeds (median contrast 0.158–0.176 against a noise envelope of 0.084–0.097), and its leading H_1 class aligns with the world's periodic coordinate (agreement 0.88–0.89, exact-belief ceiling 0.92). The activation geometry tracks the 32-symbol Bayesian belief more than the last-symbol posterior. A timing hypothesis was rejected, the matched permuted-emission control turned out to be one-sided, the last-symbol embedding alone carries 0.47–0.89 of the contrast, linear probes recover no more belief than the exact next-symbol law does, and a clean-rerun gate failed on one missing historical record. More than a dozen follow-up studies, almost all preregistered, then went after the origin of the circular structure, its causal use, and the necessity of topological distinctions for longer prediction horizons. Most stopped at preregistered feasibility gates (INCONCLUSIVE) or returned scoped negatives; positive results are exact (oracle) facts about designed processes: an orientability distinction needed only two steps ahead, component identity that requires long history, and a process whose predictions depend on the order of loop crossings. In the most direct origin test, the covariance of the learned chart phase with the part of the belief that the last four observations do not determine was excluded from beyond a ± 0.01 margin in all three models; the registered margin, however, lies above what the oracle's own phase achieves. The most recent completed study trains twelve models on a non-abelian (S_3) transport process to test whether trained transformers use the order of seam crossings beyond every commutative summary; all validity gates passed and the preregistered outcome is INCONCLUSIVE. In every seed the trained models carry a small but clearly non-zero order-dependent component on natural histories (mean $\tilde{\beta} = 0.061$, lower bound 0.0595; commutative controls and initialisations near zero), an order of magnitude below the registered effect size of 0.25, and they do not beat the bar on the designed witness histories; one commutative control's response to a designed placebo prevents the registered "order not detected" verdict. No causal claim about any trained model has been established.

Contents

1 Introduction **4**

2 Background and related work **6**

2.1 Transformers, hidden Markov models and belief states 6

2.2 World models, emergent geometry and in-context representations 6

2.3 Formal languages, automata, groups and state tracking 7

2.4 Order sensitivity versus bag-of-words 7

2.5 Mechanistic interpretability, probing and interventions 7

2.6 Grokking and learning dynamics 8

2.7 Topological data analysis of representations 8

2.8 Concentration bounds and resampling 8

2.9 Preregistration, reproducibility and leakage 8

3 The research programme **9**

4 Methodology **9**

4.1 The synthetic processes and the exact oracle . . . 9
4.2 The transformer learners . . . 10
4.3 Measurement: persistent homology and the F1 rule . . . 11
4.4 Preregistration and frozen protocols . . . 11
4.5 Dual-track planning and independent review . . . 11
4.6 Independent test authors . . . 12
4.7 Metered compute . . . 12
4.8 Registries against data leakage . . . 12
4.9 Sealing until the decision stage . . . 12
4.10 The technical-correction rules . . . 13

5 The pilot **13**
5.1 Validating the measurement on known topology . . . 13
5.2 The topological difference exists in the exact belief . . . 13
5.3 The main experiment under the smooth emission O . . . 13
5.4 The permuted control P and training dynamics . . . 14
5.5 Controls, probes, circular coordinates, sensitivity and a clean rerun . . . 14

6 Origin of the circular structure (charter priority 1) **15**
6.1 Circular-baseline audit . . . 15
6.2 How fast the circular posterior converges, and an eight-symbol cohort . . . 16
6.3 Origin beyond the last four symbols, first attempt: the frozen chart rule fails . . . 17
6.4 Origin beyond the last four symbols: does the learned phase covary with that information? . . . 17
6.5 Component identity beyond short windows (oracle only) . . . 18

7 Causal use (charter priority 2) **19**
7.1 Erasing the circular coordinate . . . 19
7.2 Counterfactual state forcing . . . 19
7.3 Readout attribution of the circle code (running) . . . 19

8 Necessity and prediction horizons (charter priority 3) **20**
8.1 Two-step-ahead distinctions in the pilot world . . . 20
8.2 An orientability distinction needed only two steps ahead (oracle) . . . 20
8.3 Orientability in trained models: do transformers carry the world posterior? . . . 20

9 Secondary programme: geometry and learning speed **21**

10 Secondary programme: non-abelian transport and crossing order **21**
10.1 Crossing order in the oracle: order-dependence certified . . . 21
10.2 Crossing order in trained models: do transformers use it? . . . 22
10.2.1 Results at the decision stage . . . 24

11 Engineering and process **25**
11.1 Coordinator, workers and the mailbox 25
11.2 The cloud training pipeline and its integrity gates 26
11.3 Defects found and corrected 27

12 Compute **27**

13 Current status and next steps **28**

14 Threats to validity and limitations **28**

15 Conclusion **29**

A Glossary **32**

B All stages, gates and outcomes **32**

C Sources **34**

1 Introduction

Large language models are trained to predict the next token, yet they are often described as carrying "world models". Whether, and in what sense, next-token training produces representations that mirror the latent structure of the process generating the data is an empirical question that is hard to answer on natural text, because the latent structure is unknown. Synthetic processes whose latent state, dynamics and exact Bayesian predictor are fully known make the question tractable. Prior work established that transformers trained on hidden Markov models (HMMs) linearly represent the geometry of the Bayesian belief state in their residual stream [67, 75], and that sequence models trained on games or walks can carry recoverable world state [14, 48, 80]. Topology—global invariants such as the number of independent loops—had not been controlled in this setting.

The **topo-pilot** project trains a small causal transformer from scratch on symbols emitted by a lazy random walk on one of two worlds with the same number of states and the same local rule: a *cylinder* (one independent loop, $\beta_1 = 1$) and a *square* ($\beta_1 = 0$). It asks three nested questions, fixed in the binding specification `docs/RESEARCH_SUMMARY.md` §1 and later in the owner's research charter (`agent-handoff/NEXT_RESEARCH_AND_WORKFLOW_EN.md`):

RQ1 Existence. Does the context-dependent representation acquire structure that reflects the latent topology?

RQ2 Origin. If so, does that structure come from the geometry of the observations, from beliefs accumulated over the history, or only from what is needed to predict the next token? (Charter priority 1: how much of the circular structure is explained by the last symbol or a short context, and how much requires long history?)

RQ3 Use. Does the model use the structure for prediction, or is it merely detectable? (Charter priority 2.) A third charter priority asks whether some topological distinctions are unnecessary for the next symbol but necessary for a longer future.

Two secondary programmes extend the question: does the world's geometry change learning speed, and can a model use topology richer than H_1, namely non-abelian transport, where the *order* in which loops are traversed matters? Table 2 in §3 lists every study by a descriptive name; the short internal labels of the project records (for example K4b or X6S1) appear only there and in the appendix tables, as keys for readers who want to look up a record in the repository.

The project is also an experiment in research process. I froze every decision rule before the data it decides existed. Each protocol was planned twice independently and reviewed result-blind before freezing, and the critical tests were written by authors who never saw the implementation. Compute is metered, fresh test histories are protected by exposure registries, and failures are classified as technical (correctable at most twice), inconclusive, or scientific. I did much of the work with language-model agents: OpenAI Codex and Anthropic Claude models served as independent planners, result-blind reviewers, implementers and test authors, and a Claude session acted as coordinator (issuing rulings and writing reviews under my authority) with workers launched per job, all communicating through a version-controlled mailbox. I approved every scientific decision point.

Summary of findings. Table 1 lists the main outcomes; Figure 1 places all of them in time.

1. **RQ1, positive under the smooth emission O.** The preregistered F1 rule is supported in every panel and replicate; the detected loop is the world's periodic circle (§5).
2. **RQ2, partially answered.** The geometry follows the full-history belief, also among histories with the same last symbol, and the learned circle exceeds an optimistic last-symbol bound. But the last-symbol embedding already carries much of the contrast, the timing

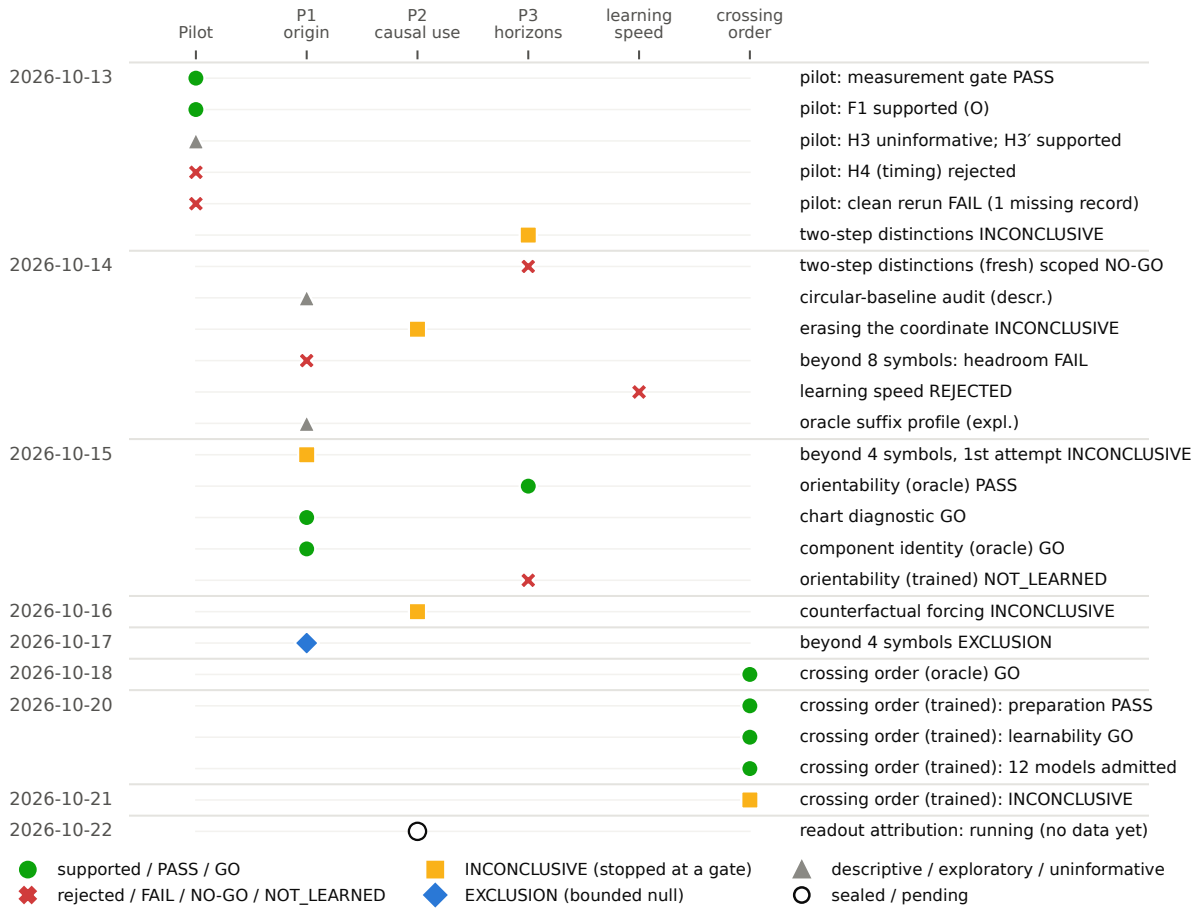


Figure 1: Programme timeline. Each row is one recorded stage outcome, placed in the column of the research question it served (P1–P3 are the owner charter's priorities). Order within a day follows the decision log; spacing is not to scale. Sources (one per row, in `make_figures.py`): `state/DECISIONS.md` D-017, D-049, D-052, D-054, D-057, D-063, D-073, D-074; `reports/E2/summary.md`, `reports/E3/summary.md`; `x4-history-feas-store/GATE_REPORT.json`; `agent-handoff/jobs/<job>/review.md` for jobs 0018, 0020, 0023, 0025, 0033, 0044, 0057, 0066, 0089, 0092, 0093, 0094, 0099 and 0107.

hypothesis was rejected, beliefs converge quickly with history length, and the most direct test found the beyond-four-symbol covariance of the learned phase to be smaller than 0.01—a margin that the oracle's own phase also does not reach (§6).

3. **RQ3, open.** Both intervention designs, erasing the circular coordinate and forcing a counterfactual state, stopped at preregistered feasibility gates before any causal endpoint was scored (§7). A third design, which attributes the model's own response to natural counterfactual histories exactly to its circle code, is running.

4. **Horizons and richer topology.** Exact oracle computations prove the existence of an orientability distinction that is invisible one step ahead and needed two steps ahead, but the trained-model study of it failed its learnability gate. An S_3 transport process was certified to carry order information beyond every commutative statistic; the trained follow-up reached its preregistered decision, INCONCLUSIVE: a weak but reproducible natural order sensitivity, far below the registered size (§8, §10, §10.2.1).

2 Background and related work

2.1 Transformers, hidden Markov models and belief states

An HMM with transition matrix T and emission matrix E induces, for every observed history, a posterior over latent states: the belief state, obtained by Bayesian filtering [69]. The set of reachable beliefs, together with the map from belief to future-symbol distributions, is the object studied by computational mechanics, where the minimal sufficient statistic of the past for the future (the causal state, or ε-machine state) is the natural notion of what an optimal predictor must remember [18, 76]. Predictive-state and observable-operator representations express the same information through future-observation statistics [42, 78]. Shai et al. showed that transformers trained on small edge-emitting HMMs linearly encode the belief-state simplex, including fractal belief geometries, in the residual stream, sometimes only in the concatenation of layers [75]; Piotrowski et al. explained the layer-wise geometry as constrained belief updates by attention followed by nonlinear correction [67]. Related accounts treat in-context learning as implicit Bayesian inference over latent concepts [82], show how transformers learn Markov-chain statistics in context through induction-like heads with stage-wise dynamics [9, 21, 60], and study what function classes can be learned in context [30]. More recent preprints report factored representations when latent factors are independent [74] and belief geometry in pretrained models in context [3].

Relation to this project. The pilot uses exactly this oracle machinery (§4.1): the exact filter supplies the 32-symbol belief $b^{(32)}$, the last-symbol posterior $b^{(1)}$ and the future laws F_1, F_2, and every learned geometry is compared with these model-free spaces on the same histories. Two design choices come directly from this literature. First, the axis-wise emission permutation P (§4.1) preserves the exact product structure of the smooth emission, because factor-wise representations are expected [74]. Second, layer-wise and concatenated hooks are analysed because belief geometry need not sit in one layer [67, 75]. The pilot's addition is a controlled *topological* contrast at matched state count, and persistent homology on clouds of individual histories rather than on per-state centroids.

2.2 World models, emergent geometry and in-context representations

Othello-playing sequence models carry recoverable board state [48]; models trained on random walks can predict well without a coherent world model by stricter metrics [80]; pretrained language models represent space and time linearly [33] and some features are irreducibly multi-dimensional, including circular ones such as days of the week [26]. Pretrained models reorganise their representations to match a graph defined in context, but the analysis there uses per-node averages, whose principal components reproduce the spectral embedding of the graph [65]. Closest to the pilot in spirit, Jerpelea planted a ring-structured latent Markov variable in generated text and found ring geometry in a from-scratch student, using centroids and PCA [43]; Brenner et al. studied fully observed lattice walkers, whose residual stream encodes running averages of steps [14].

Relation. These results make two pitfalls concrete. Per-state averages import the world's structure from outside the model, and short-range symbol mixing alone can create global lattice structure. The pilot therefore forbids centroids, UMAP/t-SNE and k-NN geodesics as evidence, and adds two model-free mixing baselines (previous-token mixing of a spectral symbol embedding, and the context average), together with the trained last-symbol embedding (`docs/RESEARCH_SUMMARY.md` §7.2, §7.6). The claim of the pilot is accordingly narrow: the general phenomenon "periodic latent dynamics can shape periodic geometry" has prior art [43]; the pilot contributes a matched two-topology comparison with these controls. I do not claim priority.

2.3 Formal languages, automata, groups and state tracking

Theoretical work bounds what fixed-depth self-attention can compute: hard and soft attention cannot robustly recognise parity or Dyck languages at arbitrary length [34], empirical studies map the formal languages transformers learn [8], and a Chomsky-hierarchy benchmark finds transformers weak at state tracking beyond regular languages without structural help [20]. Log-precision transformers lie in uniform TC^0 [55]; since the word problem of S_5 is NC^1-complete by Barrington's theorem [4], neither transformers nor common state-space models can track general non-solvable group state at arbitrary length unless $\mathrm{TC}^0 = \mathrm{NC}^1$ [56, 73]; negative eigenvalues restore some state tracking in linear RNNs [32]. For *solvable* groups, transformers learn shallow "shortcuts" whose existence follows from the Krohn–Rhodes decomposition [46, 49]. Empirical mechanistic work shows that networks learn group operations through representation-theoretic circuits [16], that language models track permutation state with associative and parity-associative mechanisms [47], and a recent preprint argues that standard transformers learn cosets of normal subgroups [85].

Relation. The crossing-order studies rest on this distinction. Their transport group S_3 is non-abelian but solvable, so the composition task is expressible at fixed depth [49], so expressivity is not in doubt; the question is whether next-token training on a *topologically generated, implicitly observed* group word yields order-sensitive predictions beyond commutative summaries. The commutative competitors form a ladder (parity, which is the quotient S_3/A_3; signed crossing counts, i.e. the H_1 class; unsigned counts; the full Parikh vector of directed crossings, the strongest commutative statistic of a word [64]). The coset account predicts either full order use or parity-level behaviour for S_3, whose normal subgroups are trivial, A_3 and S_3 [85]; the trained crossing-order protocol records this as a named competing hypothesis, checked against the preprint's abstract only.

2.4 Order sensitivity versus bag-of-words

Transformer language models use much less word-order information than their architecture permits: ablating order within long-range context changes little [62], and pretraining on shuffled text retains much downstream performance [79]. *Relation.* The crossing-order studies make the order question exact: on designed history families with identical token and directed-bigram multisets, the oracle's next-symbol law still differs, so any bag-of-tokens predictor has a certified excess loss (§10.1). On the natural distribution, however, bag mechanisms are *not* excluded by the trained study's design, and the protocol says so.

2.5 Mechanistic interpretability, probing and interventions

Circuit-level analysis of transformers [24], induction heads [63] and superposition [25] supply the vocabulary of residual stream, heads and readouts used here. Linear probes [1] measure decodability, not use; control tasks and selectivity are needed to interpret them [6, 39]. Linear concept erasure [7, 70] removes linearly decodable information but may leave nonlinearly decodable information; causal claims require interventions such as interchange interventions, causal abstraction or activation patching [31, 37, 54, 84].

Relation. The pilot's probes compare decoding of $b^{(32)}$ from activations against decoding from the exact next-symbol law and from a one-hot last symbol, which acts as a competitor-matched control in the spirit of Hewitt and Liang [39]. Both of the project's causal designs met the known failure modes of interventions: the targeted linear eraser removed linear recoverability while a nonlinear (RBF) decoder still recovered the coordinate at 0.998, and no random eraser matched its damage; the state-forcing direction shifted protected readouts that matched controls left nearly untouched (§7). The third design uses interchange of natural histories at a site where the readout

is exactly linear.

2.6 Grokking and learning dynamics

Small transformers on algorithmic tasks can generalise long after fitting the training data [68]; mechanistic progress measures reveal a gradual circuit formation behind apparently sudden transitions [59], and different runs can implement different algorithms for the same modular task [86]. *Relation.* H4 asks a timing question of this kind (does topology appear only after history use begins?), and the trained crossing-order study trains for 30,000 updates because the orientability study's 10,000-update recipe showed a flat dev loss from step 1,000; I chose the longer run; it is not a tuned value.

2.7 Topological data analysis of representations

Persistent homology tracks homology classes across a filtration [15, 22, 88]; diagrams are stable under perturbation [17], admit confidence sets [27], and circular coordinates can be extracted from persistent cohomology classes [19]. Vietoris–Rips barcodes are computed efficiently by Ripser [5]; universality of random persistence diagrams is conjectured or proved under specific conditions [10]. Persistent homology has revealed toroidal structure in grid-cell populations [29], tracked topology through deep networks [58, 71], and been applied to language-model activations, where matched Gaussian clouds produce hundreds of spurious H_1 bars and pseudoreplication must be avoided [52]; another study argues that transformer token clouds within a sentence are topologically simple [28]. Sheaves and local systems formalise local-to-global consistency [35] and have entered neural architectures [11]; local systems and holonomy are classical [36].

Relation. The pilot computes Vietoris–Rips homology over $\mathbb{F}_2$ on clouds of 512 individual histories, normalises the longest finite H_1 bar by a cloud scale, and reports a Gaussian floor as context, never as a threshold. Following Hurlbert [41] and Malhotra et al. [52], panels within a run are not treated as replicates; the decision uses sign consistency across three seeds and eight disjoint panels. The ground truth is computed from an explicit cubical complex, because the transition graph of the square has 121 independent cycles that are filled faces, not holes. The crossing-order process uses a local system (a flat S_3 connection with one non-flat face) in the *generator*; nothing is claimed about the trained model being a sheaf.

2.8 Concentration bounds and resampling

The bootstrap [23] is used throughout for descriptive intervals and several confirmatory rules, with stated nominal (not distribution-free) coverage. Where a finite-sample guarantee is needed, the trained crossing-order study uses an empirical-Bernstein bound for a bounded loss: variance-adaptive bounds of this type are due to Audibert et al. [2] and Maurer and Pontil [53], in contrast to the variance-blind Hoeffding inequality [40]; see Boucheron et al. [12] for background. The protocol derives its own version with a paired U-statistic for the variance, at a small cost in constants relative to Maurer and Pontil [53] (§10.2).

2.9 Preregistration, reproducibility and leakage

Undisclosed analytic flexibility inflates false positives [77]; preregistration separates confirmatory from exploratory analysis [57, 61]. In machine learning, variance across seeds and tooling undermines single-run claims [13, 38, 87], and reproducibility programmes have been proposed [66]. Data leakage and test contamination are a recurrent source of invalid results [44, 45, 51, 72]. *Relation.* The methods of §4 are meant as a strict response to these problems: frozen rules,

result-blind review, three-valued outcomes, exposure registries for fresh test histories, sealed validation quantities, and bitwise rerun checks.

3 The research programme

The programme grew in three phases.

Phase 1: the pilot (12–13 October 2026). A staged plan (`docs/STAGES.md`) first validated the measurement pipeline on ground truth, then established that the exact oracle spaces separate the two worlds, then trained the models and decided the two main hypotheses, then ran the permuted-emission control and the timing analysis, and finally added controls, probes, circular coordinates, sensitivity analyses and a clean rerun. The plan names these five exit points E0–E4, and §5 follows them in order. Four checkpoints required my approval. Seventeen models were trained on CPU.

Phase 2: follow-ups on origin, use and horizons (13–17 October). After the pilot I fixed the charter priorities quoted in §1. Each follow-up was a separate, frozen protocol, listed in Table 2.

Phase 3: non-abelian topology (17 October–). An exact computation certified that an S_3 transport process carries order information beyond every commutative statistic; a trained study (frozen 18 October, authorised 20 October) then trained twelve models on it, on a free cloud GPU, with all evaluation local. From 21 October a third causal-use design has been planned, frozen and implemented; its execution started on 22 October.

How the questions connect. The pilot answered RQ1 positively under O and left RQ2 split between "observation geometry" and "history", because the last-symbol embedding alone already reproduces much of the contrast. The origin studies sharpened RQ2 into a quantitative question about suffix lengths, which the oracle suffix profile showed can only be asked for short suffixes, because the circular posterior converges quickly. The causal studies went at RQ3 directly and stopped at feasibility gates; they showed that the circular information is encoded nonlinearly and that naive interventions are not specific. The horizon studies moved from a scoped negative on the pilot world to a constructed world where a topological distinction is provably needed only two steps ahead; the trained version did not learn it. The component-identity and crossing-order oracle studies looked for processes where long and order-dependent history is genuinely required, so that a trained model's behaviour can discriminate between explanations. The crossing-order study on trained models is the first trained test on such a process.

4 Methodology

4.1 The synthetic processes and the exact oracle

Worlds. A 12×12 lattice with 4-neighbourhood, $N = 144$ states. The square has no identification; the cylinder is periodic in x. Ground-truth Betti numbers are computed from an explicit cubical complex over $\mathbb{F}_2$: square $\beta = (1, 0, 0)$, cylinder $(1, 1, 0)$. The primary metric is the intrinsic flat metric $d_M((x, y), (x', y')) = \sqrt{\min(|x - x'|, 12 - |x - x'|)^2 + (y - y')^2}$ on the cylinder (without the minimum on the square).

Dynamics. A lazy walk normalised by the maximal degree,

$$T_{ij} = \frac{1 - p_{\text{stay}}}{D_{\max}} \ (i \sim j), \qquad T_{ii} = 1 - \deg(i)\,\frac{1 - p_{\text{stay}}}{D_{\max}}, \qquad p_{\text{stay}} = 0.2,\ D_{\max} = 4, \tag{1}$$

which is symmetric and doubly stochastic, so the stationary law π is uniform.

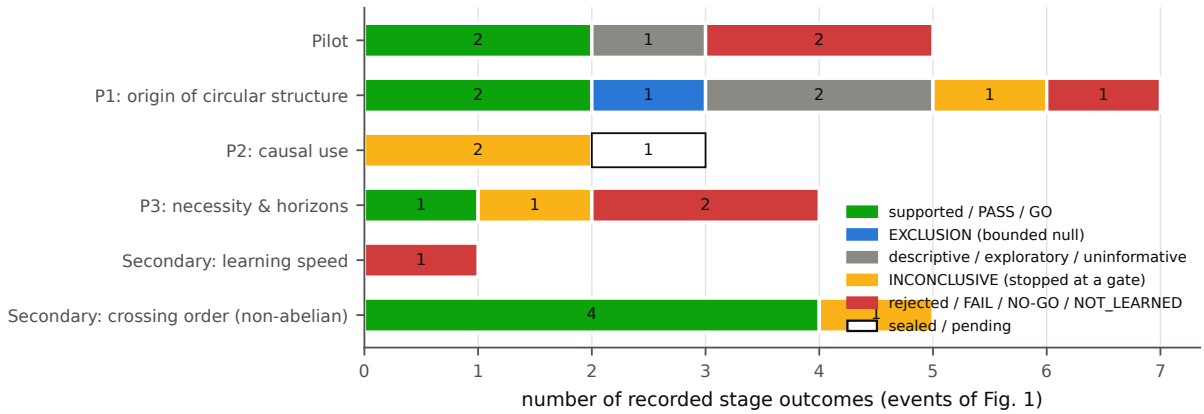


Figure 2: Outcome summary by research question: the number of recorded stage outcomes of Figure 1 in each outcome class. Supported outcomes are concentrated in the pilot and in oracle-only studies; trained follow-ups mostly stopped at gates. Source: the event list in `make_figures.py` (`EVENTS`), each with its cited review or decision record.

Emissions. Four conditions (`docs/RESEARCH_SUMMARY.md` §4):

- **O** (main): 36 symbols with centres $c_{ab} = (0.5 + 2a, 0.5 + 2b)$ on a 6×6 grid, $E_{zj} \propto \exp(-d_M(z, c_j)^2/2\sigma^2)$, symbol names randomly permuted; $\sigma = 1.25$ was the first candidate to pass the generator-only calibration rule. O factorises exactly as $E^x[x, a]\, E^y[y, b]$.
- **P** (matched control): $E_P[(x, y)] = E_O[(\sigma_x(x), \sigma_y(y))]$ with fixed axis permutations, preserving rows, entropies, symbol marginals and the product structure while destroying spatial smoothness.
- **N** (negative control): $E_{zw} = 1/36$, nothing is learnable about the state.
- **U** (sanity control): one symbol per state ($V = 144$).

Sequences and the exact predictor. $z_1 \sim \pi$; for $t = 1, \ldots, 33$ emit $w_t \sim E[z_t]$ and transition. The model sees $[\mathrm{BOS}, w_1, \ldots, w_{32}]$ and predicts w_{t+1} at every position. The filter, in row vectors and float64, is

$$a_1 = \pi, \qquad b_t = \frac{a_t \odot E_{:,w_t}}{a_t E_{:,w_t}}, \qquad a_{t+1} = b_t T, \qquad q_t = b_t T E. \tag{2}$$

From it come $b^{(32)} = b_{32}$, the last-symbol posterior $b^{(1)} = r_{w_{32}}$ with $r_w = \pi \odot E_{:,w}/(\pi E_{:,w})$, the next-symbol law $F_1 = bTE$ and the two-symbol law $F_2[i, j] = b\,T\,\mathrm{diag}(E_{:,i})\,T\,E_{:,j}$. The *context gain* measures what the full history adds to the last symbol, and the *learning gate* measures how much of it the model captures:

$$G_C = \overline{\mathrm{KL}(q_C \,\|\, q_{\mathrm{last}})}, \qquad \text{regret} = \overline{\mathrm{KL}(q_C \,\|\, p_{\mathrm{model}})}, \qquad \text{captured} = 1 - \text{regret}/G_C \;\geq\; 0.8. \tag{3}$$

Later studies reuse the same filter and sampler on other worlds: a 4×3 cylinder/Möbius seam band with a hidden world bit (the orientability studies), two 12×12 tori with a Dehn-twisted emission (the component-identity study), and a 4×4 torus carrying an S_3 local system (the crossing-order studies).

4.2 The transformer learners

Every trained model in the project uses the same pilot architecture (`docs/RESEARCH_SUMMARY.md` §6): a decoder-only pre-LN transformer [81, 83] with 4 blocks, $d = 128$, 4 heads, MLP width 512,

GELU, learned positional embeddings, final LayerNorm, untied unembedding, no dropout, about 0.8M parameters (802,304 in the crossing-order study, with 18 output symbols). Training uses AdamW [50] with learning rate 10^{-3}, 200 warm-up steps, cosine decay, weight decay 0.1, clipping 1.0 and batch 128. The pilot trained for 5,000 updates on 131,072 trajectories per replicate, with checkpoints at 0, 50, 100, 200, 400, 700, 1,000, 1,500, 2,000, 3,000, 4,000 and 5,000; seeds 1729, 2718 and 3141 change data, initialisation, symbol names and batch order together. The primary hook is `resid_post_3`, the residual stream after the last block and before the final LayerNorm. The orientability study trained 10,000 updates (with optional direct joint heads for 2- and 3-symbol futures), and the crossing-order study 30,000 updates on 2^{20} histories per model.

4.3 Measurement: persistent homology and the F1 rule

For each replicate, eight disjoint panels of 512 test histories (one per trajectory, at position 32) are shared by every hook, checkpoint and space. On each cloud a Vietoris–Rips filtration over $\mathbb{F}_2$ in dimensions 0–1 is computed, and

$$S = \max_{\text{finite } H_1 \text{ bars}} \frac{\text{death} - \text{birth}}{s}, \qquad s = q_{95}\{d(x, x') > 0\}. \tag{4}$$

For replicate r and panel i, the contrast removes what the architecture produces at initialisation and what training produces in either world:

$$D_{r,i} = (S^{\text{cyl}}_{\text{final},i} - S^{\text{cyl}}_{\text{init},i}) - (S^{\text{sq}}_{\text{final},i} - S^{\text{sq}}_{\text{init},i}), \qquad u_r = \max_c q_{95}\{|S_{c,i} - S_{c,j}| : i < j\}, \tag{5}$$

with c ranging over the four cells. **F1 is supported** iff $D_{r,i} > 0$ in at least 7 of 8 panels in each of the three replicates, $\text{median}_i D_{r,i} > u_r$ for each replicate, and every model passed the learning gate; it is rejected iff all gates passed and a condition failed; otherwise it is inconclusive. H2 uses $\Delta_{\text{geom}} = \rho_s(d_A, d_{B32}) - \rho_s(d_A, d_{B1})$ over all pairs, and again over pairs with the same w_{32}, decided by sign consistency; distances on beliefs are Hellinger. Pair correlations are never significance tests.

4.4 Preregistration and frozen protocols

The pilot's hypotheses, measures and rules were committed in `PREREGISTRATION.md` after the oracle analysis and before any training; secondary analyses were registered blind in `PREREGISTRATION_X4.md`; every operational detail of the last pilot stage was fixed before any of its results existed. A change to a closed decision required my approval at a checkpoint; the one such change was the addition of the belief-alignment hypothesis H3′ before any result under P existed. Each follow-up protocol is frozen by its SHA-256 hash; coordinator addenda may fix implementation footprint or bindings but may not change a gate, threshold, seed or decision rule. Every frozen protocol contains a threshold register, an outcome taxonomy with a licensed sentence for each outcome, and an explicit "never licensed" list. Three-valued outcomes (PASS / INCONCLUSIVE / FAIL_TECHNICAL, or the study-specific equivalents) prevent a technical failure from being read as a negative finding. Since the counterfactual-forcing study, protocols also include a *tolerance-attainability* table fixed before data (§7), and since the four-symbol origin test an *oracle-scale check* that compares margins with what the oracle itself achieves (§6).

4.5 Dual-track planning and independent review

From 14 October I had every planning task done twice, independently, under a written planning protocol: one track by Codex in a read-only sandbox and the other by a clean-context Claude subagent, with identical request files and base commits and no access to each other's output. The coordinator wrote a merge document recording agreements, conflicts, the decision on each

and what each track caught that the other missed. A third, result-blind LLM review (Codex, or a clean-context Claude reviewer when Codex was unavailable) had to return its required changes, and a confirmation pass had to return CONFIRM_FOR_FREEZE before any freeze. For example, the counterfactual-forcing protocol went through 13, then 7, then 4 required corrections; the trained crossing-order protocol through 14 required changes, including a verification of the empirical-Bernstein proof. The merge of the two counterfactual-forcing tracks found that half of one track's "random controls" exactly reproduced the primary intervention.

4.6 Independent test authors

For critical modules the tests are written from a written API contract (`API.md`) by a separate LLM author who never reads the implementation, and they are committed before the module enters the repository; the trained crossing-order study and the readout-attribution study used two independent authors. Corrections to an author's tests are made by the same author or a clean-context successor that reads only the governing ruling and the test, never module code. The implementer never writes the acceptance tests. The coordinator independently reruns the full test suite at each code freeze (e.g. 4,288 passed, 0 failed, at the code freeze of the trained crossing-order study).

4.7 Metered compute

Every scientific process runs under a meter that places the process tree in a Windows Job Object and records child CPU including exited descendants, meter CPU and a 0.1 s tail allowance per run into a store-local `cpu_ledger.json`. CPU is *measure-only* in the later studies (no CPU ceiling; a wall limit and memory limits guard against runaway jobs). The orientability training study revealed that its per-model supervisor had watched the virtual-environment launcher, not the training interpreter, so per-process CPU read about 0; the job-level meter was unaffected. A repair moved accounting to the whole process tree, and the crossing-order study's preflight checks that a deliberately over-limit child is killed with its training interpreter.

4.8 Registries against data leakage

A fresh test set is fresh only if none of its histories were ever generated, read or used for fitting in an earlier stage. From the counterfactual-forcing study onwards, each study builds an *exposure registry* of digests of every history that any earlier study could have exposed (3.36 million digests for the counterfactual-forcing study; 6,024,521 for the four-symbol origin test; about 19.7 million new ones plus the latter for the crossing-order study), and separation gates refuse a fresh pool with any collision. Reservations forbid any other history-generating job while a registry is being built. Test scratch directories inside the project root once made a registry inventory incomplete; scratch now lives outside the root.

4.9 Sealing until the decision stage

Quantities that could influence later choices are written to `sealed/` with their hashes and released only at the decision stage or at a stop. In the four-symbol origin test the validation means stayed sealed until the fresh result was recorded; in the trained crossing-order study the learnability-gate statistics, the learning curves and the headroom diagnostics were sealed, and the gate record released only PASS/FAIL/UNRESOLVED and the decision. Any model prediction on test histories before the decision stage is defined as leakage. This document follows the same rule: no model-performance number from that study was reported in earlier versions, and the numbers in §10.2.1 come only from the decision-stage report written after unsealing.

4.10 The technical-correction rules

A documented implementation or environment defect may be corrected by a coordinator ruling that names the defect, the correction commit, the affected artifacts and the evidence of independence from the outcome, with *at most two correction jobs per defect*. The failed attempt moves append-only to `superseded/<job>-a<k>/`; unaffected artifacts must regenerate bit-identically. A failure that shows a protocol statement to be wrong is not a defect: it stops the study and returns to planning. Any post-data change of a threshold or tolerance needs the owner. Environment reruns that reproduce identical values do not count as corrections, and final outcomes are never rerun.

5 The pilot

5.1 Validating the measurement on known topology

Question. Does the persistent-homology measurement detect a surface hole, as opposed to graph cycles? **Design.** A gate with thresholds fixed before any measurement: correct Betti numbers; $S_{\mathrm{cyl}} \geq 0.20$ and $S_{\mathrm{sq}} < 0.10$ on d_M; synthetic circle versus disk clouds ($M = 512$ in $\mathbb{R}^{128}$) classified correctly in at least 19 of 20 draws. **Result.** $S = 0.299$ (cylinder) and 0.037 (square) on d_M; 0.231 and 0.067 on the graph metric, although the transition graphs have 133 and 121 independent cycles; synthetic calibration 20/20 (mean circle S 0.460, maximum disk S 0.045). The gate passed (`reports/E0/summary.md`). **Lesson.** Graph cycles are not surface holes; the measurement targets the latter.

5.2 The topological difference exists in the exact belief

Question. Do the model-free spaces separate the worlds? **Result.** Under O every space separates the worlds in 8/8 panels, at 3–8 times the Gaussian floor; the full belief separates most strongly (0.671 vs 0.321). Under P the exact belief barely separates (5/8 panels), and the future spaces reverse (Figure 3). The last-symbol posterior and the emission field are identical under O and P, because P permutes states. **Interpretation.** RQ1 is decidable under O, and history strengthens the expected signature. But H3 ("the structure survives under P") became *one-sided* before any P model was trained: a failure under P would be expected even of the ideal predictor. This reading was registered before the P runs.

5.3 The main experiment under the smooth emission O

Design. Six runs (2 worlds × 3 seeds) plus one N model, 13–16 CPU minutes each. **What ran.** All six passed the learning gate (captured 0.985–0.986 for the cylinder and 0.992 for the square; regret 0.0018–0.0039 nats against $G_C = 0.2649$ / 0.2360). **Result.** F1 **supported**: $D > 0$ in 8/8 panels in each replicate, median $D = 0.158$, 0.176, 0.173 against $u_r = 0.097$, 0.084, 0.084 (Figure 4a). At initialisation the worlds are indistinguishable ($S \approx 0.06$); after training the cylinder reaches $S \approx 0.31$ and the square ≈ 0.13. H2 is sign-consistent: $\Delta_{\mathrm{geom}} > 0$ in 48/48 panels over all pairs (medians 0.087–0.127), and among same-w_{32} pairs, where the last-symbol embedding is constant, the activation distances track $b^{(32)}$ with ρ_s 0.935–0.979. The N model sits at $H(W) = \ln 36$ (NLL 3.5835, half-width 0.0002): no leakage. In a secondary analysis over six views of the final state the F1 rule holds in all six, including the component the readout cannot see (`reports/E2/summary.md`). **The last-symbol limitation.** The trained embedding of the last symbol alone reproduces 0.89, 0.47 and 0.54 of the model's median D; the model-free mixing baselines (previous-token mixing and the context average) stay at ≤ 0.20 (Figure 4b). F1 alone therefore does not show that history integration produces the hole.

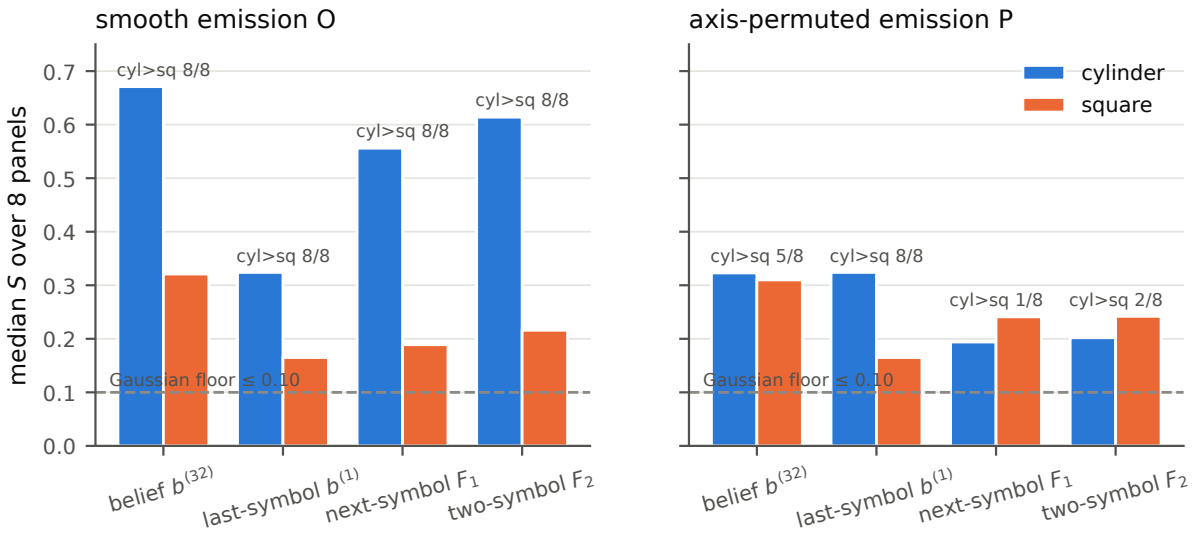


Figure 3: Oracle spaces, before any training: median S over eight panels in the exact (unlearned) spaces, replicate 1729, $\sigma = 1.25$, Hellinger distances; labels give the number of panels with $S_{\text{cyl}} > S_{\text{sq}}$. Source: `README.md`, section "Earlier exit (E1, no model)".

5.4 The permuted control P and training dynamics

P. All six P runs passed the learning gate (captured 0.874–0.914). The F1 rule was **rejected** (median D 0.021–0.034 below u_r 0.066–0.089; 6/8 and 5/8 positive panels in two replicates), which, as preregistered, is uninformative. The last-symbol-embedding contrast collapses under P (replicate 1729: 0.140 $\rightarrow$ −0.035), so its contrast under O is not static emission geometry; but whether it reflects history integration or learned symbol-level transition statistics remains open. **H3′ supported**: under P, $\rho_s(d_A, d_{B32})$ rises from 0.06–0.11 at initialisation to 0.70 and exceeds the better mixing baseline (0.39) in 48/48 run-panels. *Learned representations can align with Bayesian belief geometry even when the registered H_1 contrast is not supported*; an H_1 contrast is expected only when the belief geometry itself carries the hole.

H4 (timing). With t_h the first checkpoint where regret $< G_C$ and t_s the first where $S_t - S_0 > u_r$, all three cylinder runs under O had $t_h = 100$ and $t_s = 50$: **H4 rejected** (Figure 5). The early rise of S also occurs in the square, so it is not cylinder-specific. Exploratorily, the cylinder–square contrast on panel 1 grows mainly between steps 100 and 200 (+0.02 to +0.07 at step 50, +0.18 to +0.23 at step 200); the rule was written on raw S, not on the contrast, so this observation does not alter the verdict. A layer sweep shows the O contrast already at the embedding (≈ 0.105) and largest after the final LayerNorm (≈ 0.28), with P near zero at every hook (Figure 6).

5.5 Controls, probes, circular coordinates, sensitivity and a clean rerun

Controls. Two further N models sit at $\ln 36$ within 2×10^{-5}. The U models learn the visible-state dynamics (captured$_U$ 0.999) and also form an x-aligned circle (agreement 0.601), showing that symbol-level transition statistics alone can produce such a circle. **Probes.** Ridge probes fitted on dev panels decode $b^{(32)}$ from activations with test R^2 0.829 under O—no better than from the exact next-symbol law F_1 (0.838), far above initialisation (0.483) and the one-hot last symbol (0.421); the shuffled-label control is near zero (Figure 7a). The preregistered pattern (activations above all three competitors) held in 1/6 runs per condition. Under O the selected ridge λ was the smallest grid value in every run, so R^2 may be slightly underestimated; the grid was frozen. **Circular coordinates.** From the leading H_1 cocycle (over $\mathbb{F}_{47}$, at mid-lifetime)

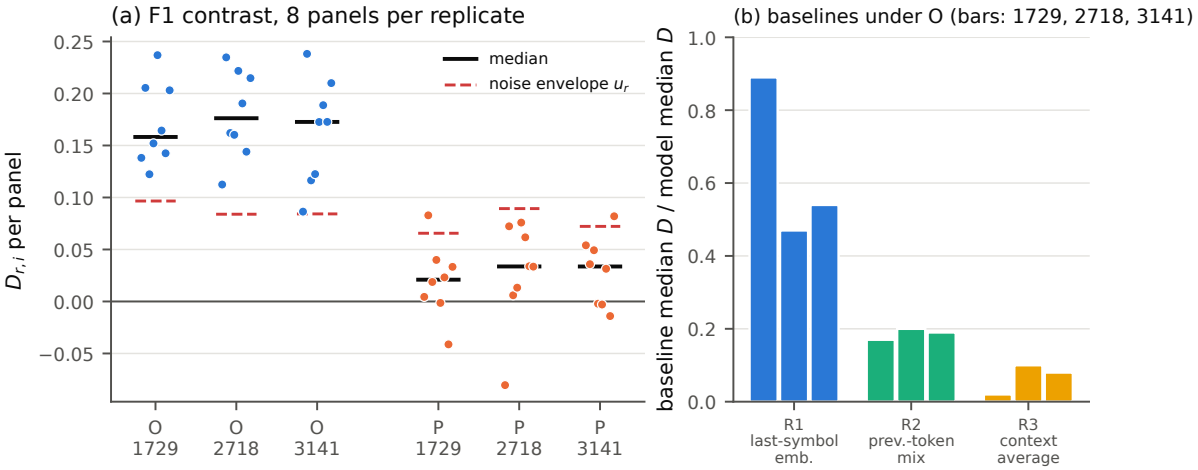


Figure 4: (a) F1 contrast $D_{r,i}$ per panel under O and P, with the median and the noise envelope u_r per replicate. (b) Fraction of the model's median D that each representation baseline reproduces under O. Sources: `reports/he_summary/data/d.json` (per-panel D, extracted read-only from the artifact store by `reports/he_summary/extract_data.py`); u_r and ratios from `reports/E2/summary.md` and `reports/E3/summary.md`.

a circular coordinate φ is computed and compared with the latent angle $\theta = 2\pi x(z_{32})/12$ by $A = \max(|\overline{e^{i(\varphi-\theta)}}|, |\overline{e^{i(\varphi+\theta)}}|)$. Cylinder models under O reach 0.884–0.892 (initialisation 0.06–0.12; non-periodic y 0.10; square 0.21–0.30; exact belief 0.915; Figure 7b). This identifies *which* loop F1 measures; it is privileged-evaluation and descriptive. A later diagnostic noted that the construction can silently give off-giant-component points an arbitrary offset; at most one point per panel was affected. **Sensitivity.** The verdicts are unchanged under the chord metric, dev-fitted PCA-32 and 256-history panels, and H3′ also under $\sqrt{\text{JSD}}$ (`reports/E4/summary.md` §6). **Clean rerun.** A fresh clone with an empty store reran the first four pipeline stages, including one full training: all 307 artifacts shared with the original store were reproduced bitwise, all six gate verdicts matched, store integrity (785/785) and report reproduction (9/9 byte-identical) passed. The clean-rerun gate nevertheless **failed**: one timing record of the original run did not exist, because the code that writes it was added after that run. The verdict stands as FAIL.

Lessons of the pilot. (i) A matched control must be evaluated on the oracle before training; without the oracle analysis, the rejection under P would have been misread. (ii) Alignment with belief geometry and a topological contrast are different properties. (iii) The wording of a rule determines what it can find: H4 was written on S, the interesting timing was in the contrast. (iv) "Bitwise reproducible" and "gate passed" are not the same statement when a gate is defined over a complete store.

6 Origin of the circular structure (charter priority 1)

6.1 Circular-baseline audit

Question. Does the learned circle exceed what the last symbol permits? **Design.** Descriptive, on the pilot's already-unblinded final test panels. **Result.** In 24/24 panels the model's agreement (medians 0.884–0.892) exceeds an optimistic last-symbol bound (0.817) and the best available baseline (0.805): outcome BEYOND_SYMBOL_AND_MIXING. **Interpretation.** This closes the pilot's missing last-symbol comparison descriptively; it is not independent confirmation and not evidence of long-history origin.

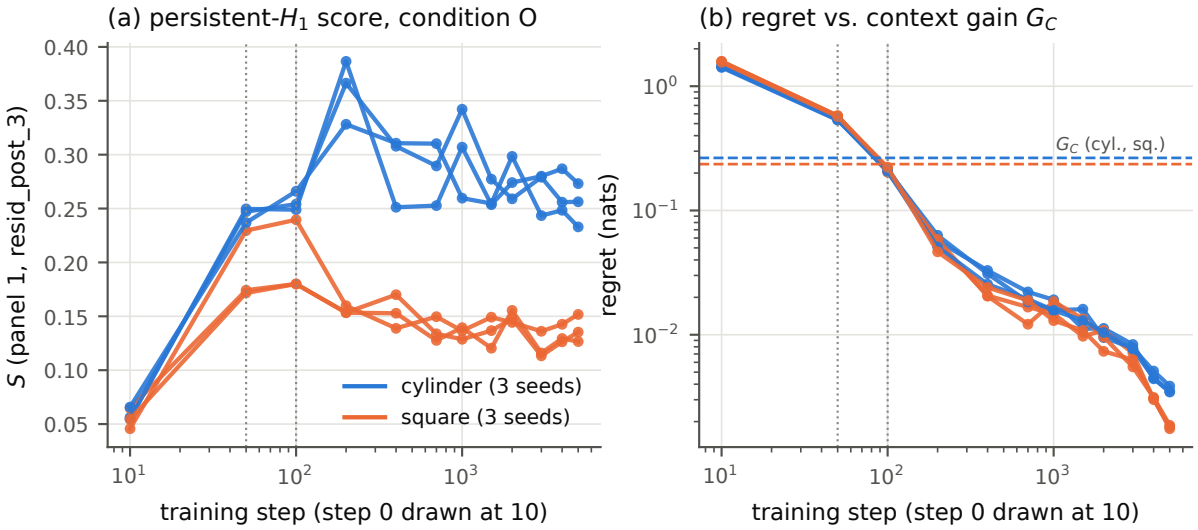


Figure 5: Training dynamics under O on panel 1: (a) the H_1 score S and (b) regret (log scale) at the twelve checkpoints, three seeds per world; dotted lines mark steps 50 and 100, dashed lines G_C. Sources: `reports/he_summary/data/dyn.json` (read-only extraction from the store, `reports/he_summary/extract_data.py`); G_C from `reports/E2/summary.md`.

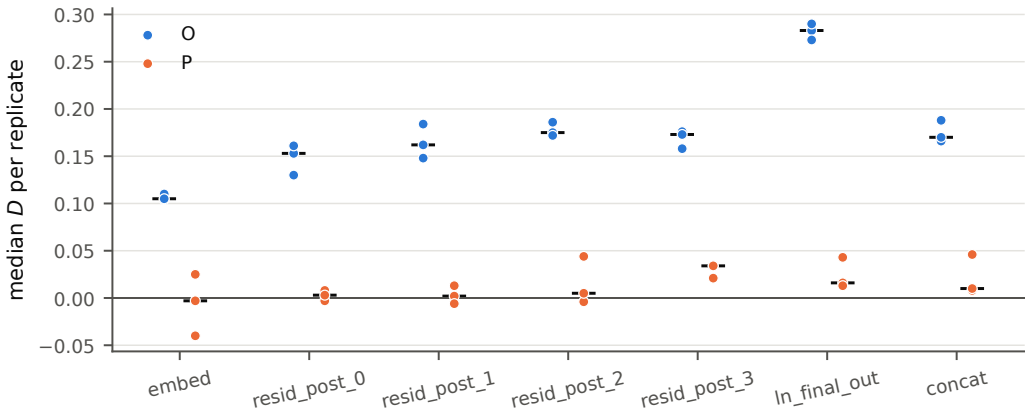


Figure 6: Layer sweep (descriptive): median D per replicate (dots) and its median (bar) for every hook, under O and P. Source: `reports/E3/summary.md`, table "Layer sweep".

6.2 How fast the circular posterior converges, and an eight-symbol cohort

Eight-symbol cohort. A preregistered cohort aimed to compare the learned phase with information beyond the last 8 symbols. Its oracle headroom gate failed: $B_8 = 0.00925$ with a one-sided 95% upper endpoint 0.00949, below the 0.01 margin. **Oracle suffix profile (exploratory).** For the first-harmonic target $m_K(h) = \mathbb{E}[c(z_{32}) \mid w_{33-K:32}]$, $c(z) = (\cos 2\pi x/12, \sin 2\pi x/12)$,

$$B_K = \mathbb{E}\,\|m_{32} - m_K\|, \qquad K \in \{1, 2, 4, 8, 16, 32\}, \tag{6}$$

was 0.369, 0.198, 0.067, 0.0092, 0.00018 and 0 on calibration (validation nearly identical; Figure 8a). All bootstrap intervals were recomputed independently. The coordinator's review corrected the worker's report: "only at $K \leq 4$" means only among evaluated grid points, and "this world cannot support memory beyond about 8" is not an accepted conclusion. **Lesson.** The circular posterior converges quickly with suffix length, so any history-origin experiment needs a small K.

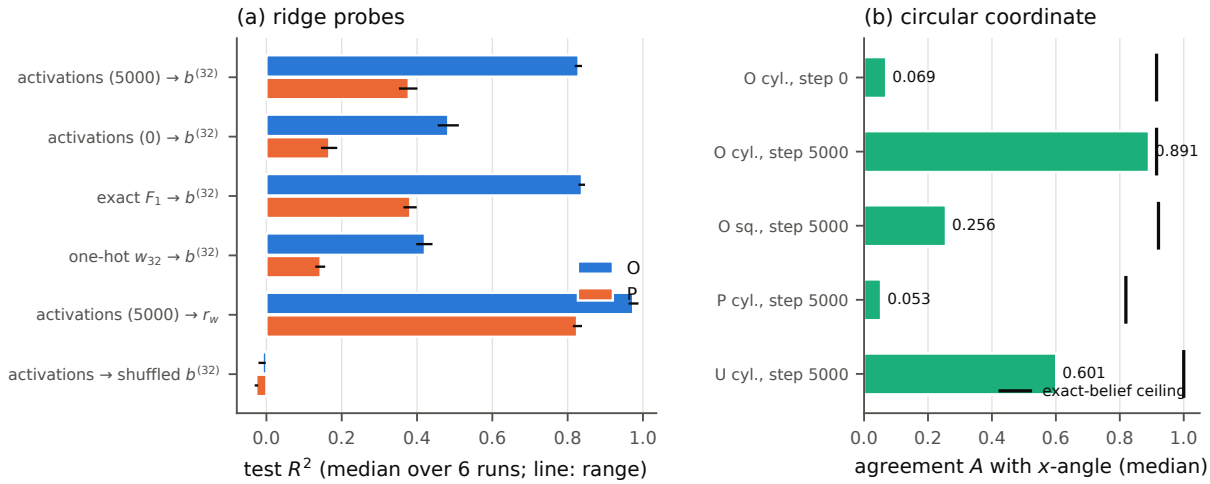


Figure 7: Probes and circular coordinates: (a) ridge-probe test R^2, median over six runs per condition with the range; (b) agreement of the circular coordinate with the latent x angle, with the exact-belief ceiling. Source: `reports/E4/summary.md` §4 and §5.

6.3 Origin beyond the last four symbols, first attempt: the frozen chart rule fails

Design. This study moved to $K = 4$ with a frozen circular-chart construction (the midpoint scale of the longest H_1 bar, landmark graph, integer lift). **What ran.** Identity and the exact process/oracle gates passed (moment errors $\leq 8 \times 10^{-16}$). **Result.** INCONCLUSIVE at chart_ calibration: at the frozen scale the landmark graph of model 1729 was disconnected; by rule, no retry, no refit and no scoring. Because the first counterfactual-forcing protocol depended on this chart by its own freeze, it was blocked at the same moment. A frozen geometry-only diagnostic then found that the disconnection was a single isolated smallest-norm point and that the largest connected component held at least 99.8% of landmarks in all six cases (GO). I approved a new rule (the largest component, gated at $\geq 95\%$) for future studies only; the first attempt stayed closed.

6.4 Origin beyond the last four symbols: does the learned phase covary with that information?

Question. In the pilot's three cylinder models under O (step 5000, `resid_post_3`, position 32), does the circular coordinate $u_j(h) = (\cos\varphi_j, \sin\varphi_j)$ built by the new chart rule on fresh landmarks covary with $r_4 = m_{32} - m_4$, the part of the first-harmonic belief that the last four observations do not determine? **Estimand.** $\Delta_4 = \mathbb{E}[u \cdot r_4]$. By the tower property $\Delta_4 = 0$ for every suffix-only coordinate. Rows without chart support contribute the interval $[-\|r_4\|, +\|r_4\|]$, so $[\mathbb{E}X^-, \mathbb{E}X^+]$ is exactly the identified set over all completions. **Decision.** Per model, SUPPORT iff $L(\mathbb{E}X^-) - F > 0.01$; EXCLUDE iff $U(\mathbb{E}X^+) + F < 0.01$ and $L(\mathbb{E}X^-) - F > -0.01$, with one-sided bounds at 0.05/3 and a numerical margin $F = \max(B_{\text{num}}, 10^{-9}) + Z_{\text{bound}}$ that makes the claim concern the documented law. **What ran.** The calibration and validation stage needed three technical corrections, each within the two-per-defect rule: a dictionary passed to `Fraction`, a byte-order mark and a lazy sequence in the registry handoff, and a missing reader for the store's own superseded digests; no outcome-bearing quantity existed at any failure point. Validation then passed every gate (support 3072/3072 in every model, oracle-headroom lower bound 0.0636) and fixed $N = 3072$ test rows. **Result.** On the fresh test, **EXCLUDE in all three models**: $[L, U]$ = [0.00640, 0.00777], [0.00740, 0.00891], [0.00667, 0.00823] (Figure 8b). Suffix-only nulls were near zero, as designed.

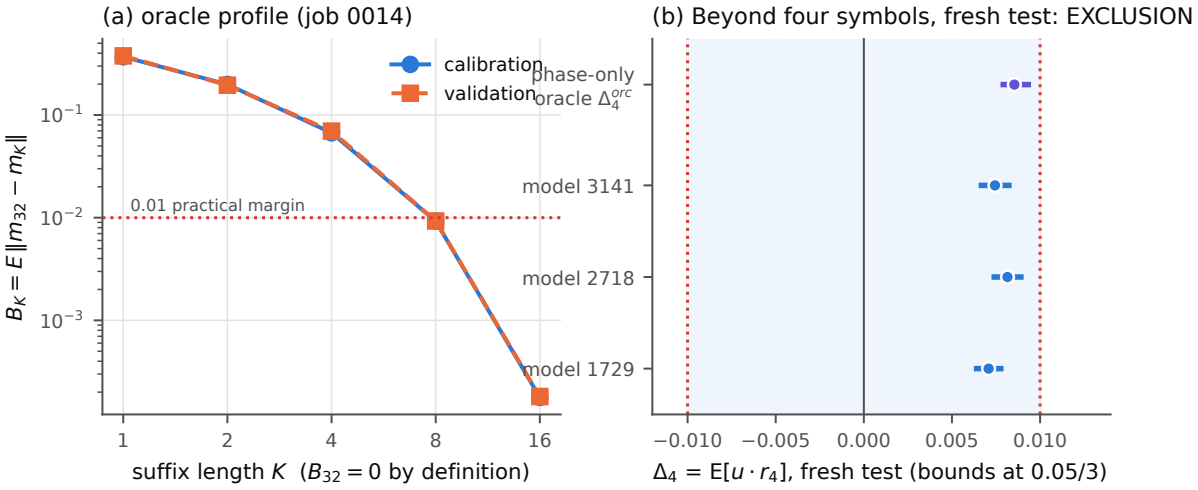


Figure 8: (a) Oracle profile B_K for the first-harmonic circular posterior (calibration and validation); $B_{32} = 0$ by definition. (b) Four-symbol origin test, fresh-test result: per model the nominal one-sided bounds $[L, U]$ (rank-84 and rank-4,917 bootstrap order statistics) and point estimate of Δ_4, the EXCLUSION margin ± 0.01 (dotted), and the phase-only oracle comparator Δ_4^{orc}. Sources: `agent-handoff/jobs/0014/review.md`; `agent-handoff/jobs/0057/review.md`.

Interpretation and caveat. The licensed sentence is that, for this chart rule, site and process, a covariance of either sign at or beyond 0.01 is excluded. But the phase-only oracle comparator itself achieves only $\Delta_4^{orc} = 0.00854$ [0.00791, 0.00933], below the margin, and the models reach 83–95% of it; all lower bounds are positive. EXCLUSION therefore means only that the covariance is small in absolute terms; it does not show that the learned phase lacks beyond-four-symbol information. The protocol registered no "positive but below margin" status, so no stronger claim is licensed. **Lesson.** Margins must be checked against the oracle's own achievable value before freezing; the trained crossing-order and readout-attribution protocols adopted this as an oracle-scale check.

6.5 Component identity beyond short windows (oracle only)

Question. In a process whose components have identical single-symbol statistics, how much component information is available from short windows versus the full history, and is the full history more than a product of block likelihoods? **Design.** Two 12×12 tori with the pilot walk; component B emits A's field pulled back by the Dehn twist $\phi(x, y) = (x + y \bmod 12, y)$; exact enumeration for windows up to 4 and 2^{20} validated Monte Carlo sequences otherwise. **Result.** GO (full history needed, non-additively): the last symbol carries no component information; window information rises from 0.002 nats (2 symbols) to 0.067 (8) and 0.325 (full history); full-history Bayes accuracy exceeds the 8-window accuracy by 0.191 (99% interval [0.190, 0.192]) and a four-block likelihood product by 0.0567 [0.0562, 0.0571], above the 0.03 threshold; a local-noise control was nearly additive (0.008). **Interpretation.** GO licenses *designing* a trained stage only, about component identity rather than the circular coordinate; short windows are not blind. It motivated the search for a construction where short windows are *exactly* blind, which became the crossing-order studies (§10).

7 Causal use (charter priority 2)

7.1 Erasing the circular coordinate

Design. Erase the direction carrying the coordinate m_{32} at the final site and compare the damage to prediction with random erasers matched in damage. **Result.** The targeted eraser removed linear recoverability, but a fixed RBF decoder still recovered the coordinate at 0.998 (LINEAR_COMPONENT_ONLY), and 0 of 10,000 random erasers fell inside the frozen damage-matching window (FAMILY_UNMATCHED). The causal endpoint was not run: INCONCLUSIVE (`docs/proposals/X4F_COORDINATOR_REVIEW_ADDENDUM.md`). **Lesson.** The circular information is nonlinearly and redundantly encoded; linear erasure cannot test its use.

7.2 Counterfactual state forcing

Question. At `resid_post_3`, position 32, of the pilot's cylinder models under O, where only the final LayerNorm and unembedding follow, does the readout map a change of the learned chart phase into a change of the predictive distribution in the direction an exact counterfactual oracle prescribes? Because the unembedding has rank at most 35, the residual-to-output Jacobian has a kernel of dimension at least 93, so the test is not trivial. **Design.** Donors $H_{\pm}$ shift the oracle state along the circle; a Frisch–Waugh–Lovell phase map fitted on calibration rows defines the forced state; 32 Haar-random whitened 2-frames form the matched control family; gates bound damage to the decoded last symbol $\hat{w}$ and the decoded y-marginal, support, registration of the counterfactual, a positive control, and a numerical certificate of every oracle row. The first version was blocked by the chart failure of §6; a revised version (C1′) replaced that dependency with the largest-component chart rule (33 anchored edits). **What ran.** The oracle stage first failed technically (an integer/string key mismatch), then passed every gate on bit-identical histories: the exact belief predicts the forced-state effect with a one-sided lower bound of 1.12 nats of mean gain, far above 0.01. The calibration stage first failed at a ridge-equivalence self-check; a coordinator ruling relaxed that tolerance after oracle-stage data existed, and the rerun reached the damage gate. **Result.** INCONCLUSIVE (no matched control family) at the calibration damage gate in all three models. The primary intervention flipped the decoded w_{32} on 29–39% of rows ($D_w(T) =$ 0.311, 0.389, 0.286 against a ceiling of 0.02) and shifted the decoded y-marginal by 2.5–3.2 times its calibration variance (ceiling 0.05), while the matched controls barely moved $\hat{w}$ (< 0.001); only 6, 0 and 4 of 32 frames matched, against 28 required. The coordinator found no implementation defect: the gap is a property of the fitted direction, not of displacement size. **Governance flag.** That ruling changed a tolerance after data existed, without my prior approval, in tension with the protocol's own rules. The damage-gate values do not depend on it (charts and earlier gate values are bit-identical), and under a strict reading the outcome would have been FAIL_TECHNICAL at the self-check, which licenses the same "assay infeasible as specified". I ratified the ruling afterwards. Since then every protocol carries a tolerance-attainability table fixed before data.

Interpretation for RQ3. No causal endpoint has been scored. Both failures tell me about the instruments and nothing about use: the information is nonlinear, and the obvious linear intervention direction is entangled with the last-symbol readout.

7.3 Readout attribution of the circle code (running)

Question. For natural pairs of histories that differ in their x-history but share the last symbol, what share of the model's own oracle-aligned change in next-symbol prediction is carried by an interchange of its registered circle code at the final position? **Design.** After the final LayerNorm the head is exactly linear, so this attribution is exact and needs no synthetic state passed through a nonlinearity, which removes the failure mode of the forcing study at its root. The code is the

6-dimensional span of the first three harmonics of the learned chart phase, made nuisance-excluding by a generalised-least-squares projector; a secondary family repeats the interchange at layer 2. The share ψ is declared USED if its lower bound exceeds 0.50 and NOT_USED if its upper bound is below 0.10 together with a bound on the total effect; claims need two of the three models. **Status.** After two independent plans, a comparison, a merge and three rounds of result-blind review I froze the protocol on 21 October, revised it once before any data existed (a translation placebo on the y-axis replaced a reflection that failed on this process), and authorised execution on 22 October. The first stage has not yet produced data: its first three attempts stopped at entry checks before anything was sampled (a release-record field name; a lineage check that refused the stage's own entry records; then the same check refusing the compute meter's own lock file). Each defect was corrected under the two-per-defect rule, and the stage is being rerun.

8 Necessity and prediction horizons (charter priority 3)

8.1 Two-step-ahead distinctions in the pilot world

Question. Are there reachable beliefs that agree in F_1 but differ in F_2, so that an ideal predictor must represent more than the next-symbol law? **Result.** The ambient tangent ranks allow it ($Q_1 = 35$, $Q_2 = 137$), but the four old pools were undercovered (49–66 eligible anchors against 128 required): INCONCLUSIVE. Four fresh pools (159/141/159/156 anchors) gave **zero** witnesses: a scoped NO-GO for the practical horizon gap at those thresholds and that resolution. It is not a universal sufficiency theorem; the planned next phase was not run.

8.2 An orientability distinction needed only two steps ahead (oracle)

Design. A minimal 4×3 seam band, cylinder or Möbius, with a hidden world bit Z (prior $\frac{1}{2}$); five tokens $\sigma_0, \sigma_2, \sigma_3, A, B$. Theorems in the frozen protocol: at every decision position ($w_t = \sigma_3$) the next-symbol law is identical in both worlds; at horizon 2 the world-conditional laws differ only in four entries. **What ran.** Exact rational enumeration of all 37,190 reachable histories up to length 8, a float oracle agreeing to 4×10^{-15}, and an 80-digit Decimal reference on 650 histories. **Result.** PASS: every pre-stated exact value was reproduced, including the posterior-isolation pair (identical location beliefs, world posteriors $\frac{1}{4}$ vs $\frac{3}{4}$, identical next-symbol law, two-step TV $\frac{1}{100}$). **Licence.** An existence result about an orientability distinction (equal Betti numbers) at reachable histories; nothing about the square and cylinder, β_1, or trained models.

8.3 Orientability in trained models: do transformers carry the world posterior?

Question. At decision positions, how much of $\pi(h) = P(Z = M \mid h)$ is recoverable from the activation, and how does it depend on the training objective horizon $H \in \{1, 2, 3\}$ and on the in-context decision position $t \in \{16, 28\}$? **Design.** 4 paired seeds $\times$ 3 horizons plus 2 null models (14 models, 10,000 updates); two Bonferroni-controlled confirmatory contrasts; a 2-model learnability gate first. At band-edge positions the oracle contrast $d_i = 2(a/4)(\pi - \frac{1}{2})(\delta_M - \delta_C)$ is compared with the model's $r_i = [\hat{q}(A) - \hat{q}(B)] - [Q_1^{1/2}(A) - Q_1^{1/2}(B)]$ by

$$\beta_1 = \frac{\sum_i r_i d_i}{\sum_i d_i^2}, \qquad \kappa_1 = \frac{\beta_1 - \beta_1^{(8)}}{1 - \beta_1^{(8)}}, \tag{7}$$

and learned(m) requires $\beta_1 \geq 0.30$, its lower bound ≥ 0.15, $\kappa_1 \geq 0.20$ and a positive paired lower bound of $\beta_1 - \beta_1^{(8)}$. **Result.** NOT_LEARNED: both gate models (objective horizon $H = 3$) are defined FAILs (β_1 0.393 and 0.262, κ_1 -0.890 and -1.301), below the exact window-8 predictor's $\beta_1^{(8)} = 0.679$ (Figure 9); the remaining 12 models were not trained. Dev cross-entropy was flat

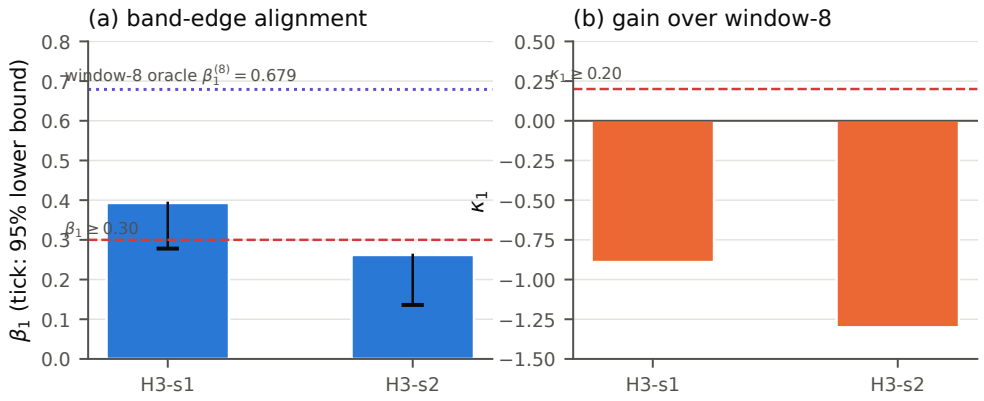


Figure 9: Orientability in trained models, learnability gate on validation: (a) band-edge alignment β_1 with its 95% lower bound against the threshold 0.30 and the exact window-8 value; (b) the normalised gain κ_1 against 0.20. Source: `agent-handoff/jobs/0033/review.md`.

at about 1.12 from step 1,000. **Licence.** "This recipe failed the registered learnability gate"—a scoped feasibility result that says nothing negative about representations.

9 Secondary programme: geometry and learning speed

Question. Does a metric-coupled "conductance" kernel (horizontal edge lengths $1+0.2\cos(2\pi y/11)$, conductances $1/l^2$) delay a grid-defined regret-threshold event by at least 250 steps relative to the flat kernel, in each of three paired models, on the same cylinder and emission? **Design.** 27-point step grid; PASS when mean regret $\leq 5\%$ of G_C at three consecutive grid points; paired bootstrap bounds (`docs/proposals/X5_G1_TRAINING_PROTOCOL_V4.md`). **Result.** Paired delays 250, 0 and 0 steps; one pair's upper bound is $0 < 250$, so the hypothesis is **REJECTED** on dev, with no censoring; the programme was closed without a fresh test. **Interpretation.** The kernel change is confounded with movement speed (the feasibility gate showed a large speed difference), so this is not a curvature null. A mean-rate-matched successor is a listed but unscheduled option.

10 Secondary programme: non-abelian transport and crossing order

10.1 Crossing order in the oracle: order-dependence certified

Construction. A 4×4 torus with the pilot's lazy walk (each move and stay with probability $1/5$). Each vertex emits its own token except an anchor $a_\star = (2, 2)$, which emits a cue revealing a hidden fibre value $x \in \{0, 1, 2\}$. Crossing the x-seam applies the transposition $a = (01)$ to the fibre, crossing the y-seam applies $b = (12)$; every face is flat except one corner face with a 3-cycle flux, and the fibre is redrawn at rate $\eta = 1/50$ (a 48-state lift). For a segment γ since the last cue, the holonomy is $\rho(\gamma) = g_{e_n} \circ \cdots \circ g_{e_1}$ and the Parikh vector $\kappa(\gamma) = (k_{x^+}, k_{x^-}, k_{y^+}, k_{y^-})$ counts directed seam crossings. Two abelian control processes, AB (a on both seams) and ZC (3-cycles), have the same counts but no order effect. **Outcome criteria.** (i) Designed: on designed witness families the cue-normalised TV between members of different holonomy is ≥ 0.3 while every commutative competitor is constant; (ii) natural: the natural excess log-loss of the best Parikh predictor $G_{\text{Parikh}}(32) \geq 10^{-3}$ nats and the fraction M_{32} of histories with row-KL ≥ 0.01 is $\geq 1/32$, both as certified enclosures. **What ran.** Two technical corrections (a key accumulation in one gate and a sign error in another gate's estimator), each confirmed by the next run. **Result.** GO,

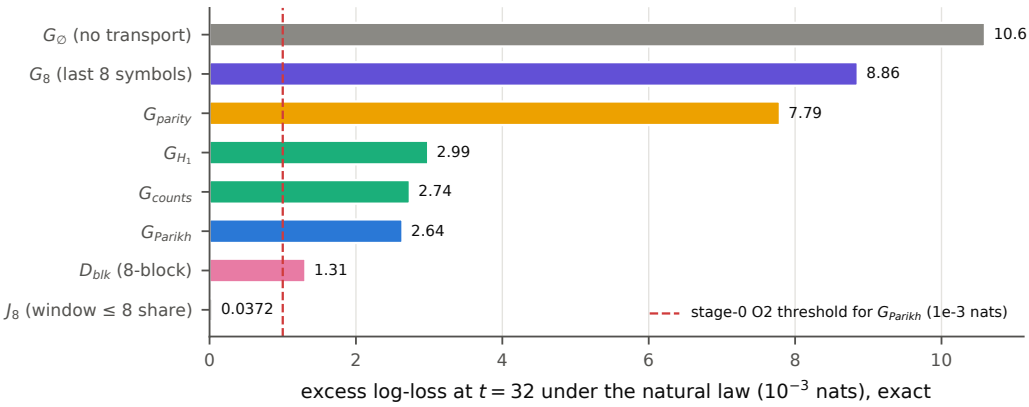


Figure 10: Crossing order in the oracle: exact natural excess log-loss at $t = 32$ of the Bayes predictor restricted to each commutative statistic (no transport; last 8 symbols; parity; H_1 class; unsigned counts; Parikh vector), the 8-block comparator D_{blk}, and the share J_8 of the gap carried by windows of at most 8 symbols; dashed line: the natural-criterion threshold. Every function of a statistic has at least that statistic's excess. Source: the certified stage-0 values quoted in `agent-handoff/planning/responses/X6S1-FROZEN-v1.md` §2.1, from `agent-handoff/jobs/0066/review.md` and `x6-stage0-oracle-store/run/report.json`.

with the natural criterion met: witness TVs 0.851, 0.695, 0.524; exact zeros in AB, ZC and the controls; $G_{\mathrm{Parikh}}(32) = 2.636 \times 10^{-3}$ nats (enclosure width $\sim 10^{-60}$) and $M_{32} = 0.0468$; no window of at most 7 symbols carries order information (exact). The ladder of commutative competitors is in Figure 10. **Licence.** Oracle facts about one designed process; a GO supports designing a trained stage, it does not authorise training. A descriptive note: the literal product-of-experts comparator has a transport-independent component (about 19 standard errors above zero), so it is not used as an order signal in the trained study.

10.2 Crossing order in trained models: do transformers use it?

Question. Do small causal transformers trained only by next-symbol prediction on natural histories of the non-abelian process (arm NA) make predictions that depend on the order of seam crossings, beyond the Parikh statistic and its coarsenings? The claim is behavioural; no mechanism is claimed.

Design (Figure 11). Arms NA, AB and ZC × seeds 1–4: 12 models, each trained for 30,000 updates on 2^{20} fresh histories with the pilot architecture and recipe; for each seed every arm starts from the same initial state and uses the same batch-index sequence. Training runs on the Colab free-tier GPU in resumable 1,000-update chunks (model, optimiser, scheduler and RNG state); every outcome-bearing evaluation runs locally on CPU in float64 from hashed weights. Training came in three waves: a smoke wave, a first wave (NA-s1, NA-s2, ZC-s1) and a second wave (the other nine, released only after the learnability gate said GO). The local stages prepared pools, registries, oracle and scoring checks and the training bundles; ingested each wave with integrity checks, the learnability gate and learning curves; and finally evaluated, decided and unsealed.

Primary estimand. Each eligible test history h (a cue present, age $n \geq 8$, end vertex next to the anchor) is paired with an exact uniform draw h' from its own Parikh class (same prefix, cue, age, end vertex and directed crossing counts). With $\Delta\hat{q} = \hat{p}_c(h) - \hat{p}_c(h')$ and $\Delta q^* = q^*_c(h) - q^*_c(h')$ on the cue-normalised laws,

$$\beta_{\mathrm{ord}}(m) = \frac{\sum_i \langle \Delta\hat{q}_i, \Delta q^*_i \rangle}{\sum_i \|\Delta q^*_i\|^2}. \tag{8}$$

Every function of the Parikh key scores exactly 0 (the pair shares the key), every function of the cue-free segment—including every function of the last 8 symbols—scores 0 in expectation, and the oracle

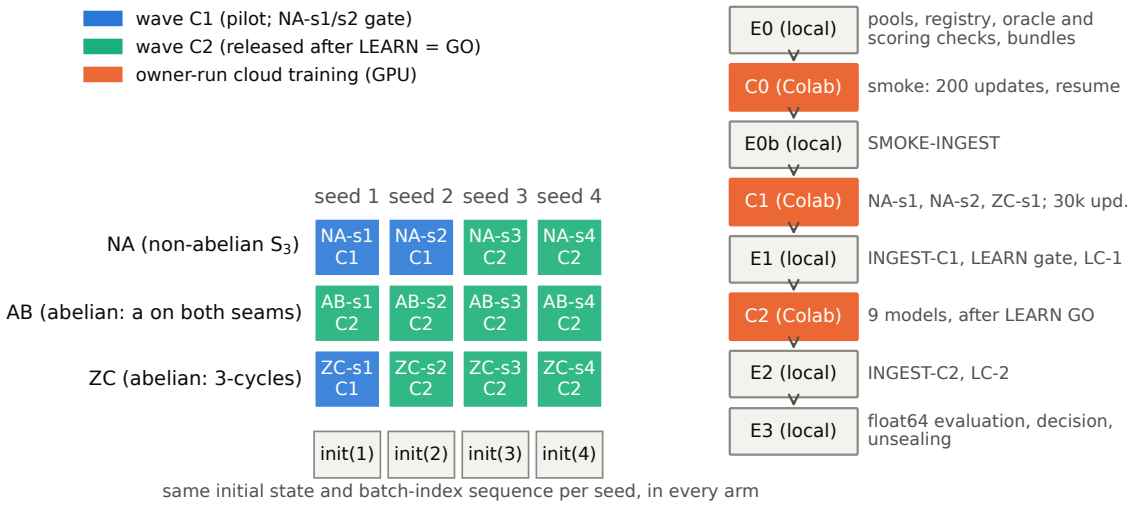


Figure 11: Crossing order in trained models, design: three arms × four seeds sharing four initial states, two training waves, and the alternation of local and cloud stages that I ran. The figure uses the protocol's stage labels (C0–C2 for the smoke, first and second training waves; E0–E3 for the local stages). Source: the frozen protocol §4.1–§4.4, §5, §8; wave parameters (smoke wave: 200 steps, $K = 100$; first and second waves: 30,000 steps, $K = 1{,}000$) from `x6-stage1-store/stages/E0/frozen_C*.json` (bundle id, steps and K only), via `docs/research_summary/data/x6s1_status.json`.

scores 1. The control-adjusted slope is $\tilde{\beta}(s) = \beta_{\mathrm{ord}}(\text{NA-}s) - \max\{\beta_{\mathrm{ord}}(\mathrm{init}(s)), \overline{\beta}_{\mathrm{ord}}(\mathrm{AB}), \overline{\beta}_{\mathrm{ord}}(\mathrm{ZC})\}$, and **C-ORD** is supported iff $\mathrm{mean}_s \tilde{\beta}(s) \geq 0.25$, its 97.5% bootstrap lower bound exceeds 0.05, $\tilde{\beta}(s) > 0$ in at least 3 of 4 seeds, and a tail diagnostic passes.

Log-loss claims and the empirical-Bernstein bound. For the quantised, ε-smoothed readout ($Q = 2^{20}$, $\varepsilon = 10^{-6}$) the per-row excess loss $X_i = \mathrm{KL}(q^*(H_i) \,\|\, p^Q(H_i))$ lies in $[0, B]$ with $B = 17$. With $n = 2m = 2^{21}$ rows, $\bar{X} = \frac{1}{n}\sum X_i$, $Z_k = (X_{2k-1} - X_{2k})^2/2$, $\bar{Z} = \frac{1}{m}\sum Z_k$, $L = \ln(2/\delta)$, $a = B\sqrt{L/m}$, $b = B^2 L/(12m)$ and $\sigma_{\mathrm{up}} = (a + \sqrt{a^2 + 4(\bar{Z} + b)})/2$, the protocol proves

$$P\Big(\mu \leq \bar{X} + \sigma_{\mathrm{up}}\sqrt{2L/n} + \frac{BL}{3n}\Big) \geq 1 - \delta \tag{9}$$

from a one-sided Bernstein moment bound and a paired U-statistic for the variance, and computes the bound outward in exact and directed-rounding arithmetic. **SHARP**(s) holds iff the bound at $\delta = 0.025/4$ is below the stage-0 lower endpoint of $G_{\mathrm{Parikh}}(32)$: then model NA-s is not a function of the Parikh statistic or any coarser one. A secondary TARGET requires recovering 25% of that gap. Designed criteria use the bag- and bigram-matched AB-versus-BA family (3,090 pairs).

Learnability gate. Before any order statistic exists, an NA pilot model must have natural excess log-loss credibly below the exact window-8 gap $G_8(32) = 8.855 \times 10^{-3}$ nats on fresh validation rows (bootstrap upper bound at 97.5%). Passing means "better than every function of the last 8 symbols"; it does not mean order use. GO iff NA-s1 or NA-s2 passes. Only statuses and the decision are released; the statistics are sealed.

What ran (Figure 13).

- *Implementation*: an API contract and two independent test authors, eleven modules within line limits set by coordinator addenda (footprint only; no scientific change), and a coordinator full suite of 4,288 passed at the code freeze.
- *Authorisation*: I authorised the frozen protocol by its SHA-256 hash, a local budget of about 43 CPU-h (forecast range 23–70), and cloud use of about 1–4 GPU-h.
- *Preparation, attempt 1*: stopped at its first lineage check; nothing scientific ran. Two defects:

a constant path one level short of its value, and a lineage check that refused the authorisation record written by the same run. Both were corrected.

- *Preparation, attempt 2*: all ten gates passed, from authorisation through pools, registries, oracle and scoring checks to the cloud bundles, and the bundles were built.
- *Smoke wave*: I ran it in Colab. The first ingest was INVALID at the provenance check, because a hash pasted with a line break produced a notebook cell that did not compile but was still recorded, and the check treated every recorded source as executed. The fix skips sources that do not compile, which cannot have executed anything; the second ingest on the same snapshot passed. One recorded open defect is that no bundle re-issue mechanism exists for a retrain; it is not needed unless a retrain occurs.
- *First wave*: the three models passed all four integrity checks, and the learnability gate **passed with the released decision GO** (both NA pilot models passed).
- *Second wave*: I ran it; the nine models passed all four integrity checks, so all 12 models were admitted.
- *Evaluation and decision*: the first evaluation attempt was INVALID because every model chunk exceeded the 3 GiB per-process memory limit (a full-pool vectorised forward peaked at about 3.0 GiB against a forecast of 1.2 GB). The correction evaluates in row blocks of 2^{16} with byte-identical outputs (peak 1.11 GiB) and was re-frozen; the regenerated initialisation chunk was bit-identical. All 16 predictor chunks then completed, and the decision stage passed every step and unsealed.

Total local compute was 79.55 CPU-h, about 21 CPU-h of it in the failed attempt (Figure 14b); I extended the budget beyond the approved 23–70 CPU-h. Cloud training took one Colab session per wave, about 10 s (smoke), 24 min (first wave) and 69 min (second wave) by the session records.

10.2.1 Results at the decision stage

All numbers in this subsection are from the decision-stage report written after unsealing (`x6-stage1-store/stages/E3z/report.json`, sha256 `90be2081...`) and the coordinator's review of it. Figure 12 summarises them.

Outcome. INCONCLUSIVE under the protocol's ninth decision rule, with the licence "descriptive only". The registered qualifiers record that one control limited the designed comparison in seed 1, that both control arms learned in all four seeds, that the finest commutative statistic the NA models beat is parity, that one designed criterion is unresolved, and that a length profile is reported. Every validity gate passed, all 12 models were admitted and learned (the learnability gate said GO, and all twelve passed the test-time learnability check), the natural placebos were clean, the tail diagnostic passed and precision was adequate in every seed.

Natural order use. On the informative natural partner pairs at $t = 32$ (the oracle has $\beta_{\mathrm{ord}} = 1$, the Parikh-class predictor 0), the four NA models reach $\beta_{\mathrm{ord}} = 0.047,\ 0.090,\ 0.072,\ 0.049$, the AB controls 0.001–0.004, the ZC controls 0.001–0.008 and the initialisations about 0. The per-seed contrasts are $C_s = 0.044,\ 0.088,\ 0.064,\ 0.045$, each with a 95% interval far from zero, and the mean is $\tilde{\beta} = 0.061$ with $\mathrm{LB}_{0.975} = 0.0595$. The registered C-ORD claim requires a mean of at least 0.25, so C-ORD is *not supported*; every seed falls in the registered "low" band ($\tilde{\beta}(s) < 0.10$) and none in the "order" band.

Designed order use. On the designed witness family W2 the NA models' regret is $R_{W2} \approx$ 0.080–0.087, above the bar $(3/4)\Delta^{\mathrm{lo}}_{W2} \approx 0.0458$, so the designed order criterion fails in seeds 2–4. Seed 1 is unresolved because the commutative control ZC-s1 responded to the designed W2 contrast (placebo $\beta_{W2} = 0.209$; other controls ≤ 0.006 in magnitude). Descriptively, the ZC controls reach a W2 regret similar to the NA models' (0.077–0.094) while the AB controls stay

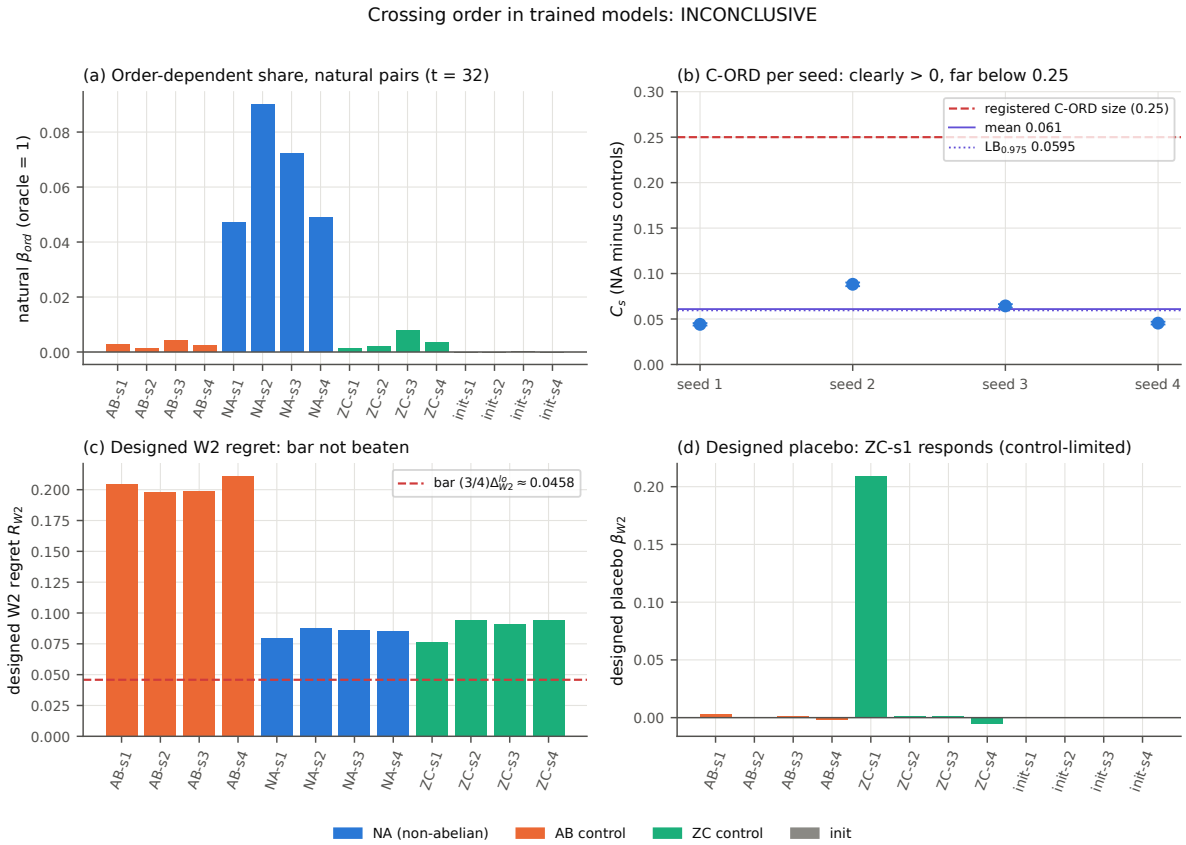


Figure 12: Crossing order in trained models, decision stage. (a) Natural β_{ord} per predictor on the informative $t = 32$ partner pairs. (b) Per-seed C-ORD contrast C_s with 95% intervals, the mean and $\text{LB}_{0.975}$, against the registered effect size 0.25. (c) Designed W2 regret against the bar $(3/4)\Delta^{\text{lo}}_{W2}$. (d) Designed placebo β_{W2} of the controls (ZC-s1 fails). Source: `x6-stage1-store/stages/E3z/report.json` via the snapshot `docs/research_summary/data/x6s1_result.json`.

near 0.20, so the designed regret does not separate order-sensitive from commutative models. SHARP and TARGET were certified for no NA model, and the finest commutative ladder rung the NA models beat is parity.

Why INCONCLUSIVE. The "order not detected" rule needs the designed criterion resolved in all four seeds. Every other condition of that rule holds; the single designed placebo failure of ZC-s1 leaves seed 1 unresolved, and the rules are applied as registered, without reclassification.

Reading. Within the licence: next-token training on natural non-abelian transport histories produced a weak but reproducible sensitivity to the order of seam crossings, present in every seed and absent from the commutative controls, at about one sixteenth of the oracle's order information. It falls far short of the order use the study was designed to detect, and the designed probes do not show it. Whether more training, larger models or a different curriculum would close the gap is a new question.

11 Engineering and process

11.1 Coordinator, workers and the mailbox

I organised execution around a separate git repository, `agent-handoff`, holding the protocol (`PROTOCOL.md`), one directory per job (`task.md`, `status.json`, `report.md`, `review.md`), the request and control files, planning requests and responses, owner decisions (immutable files plus numbered

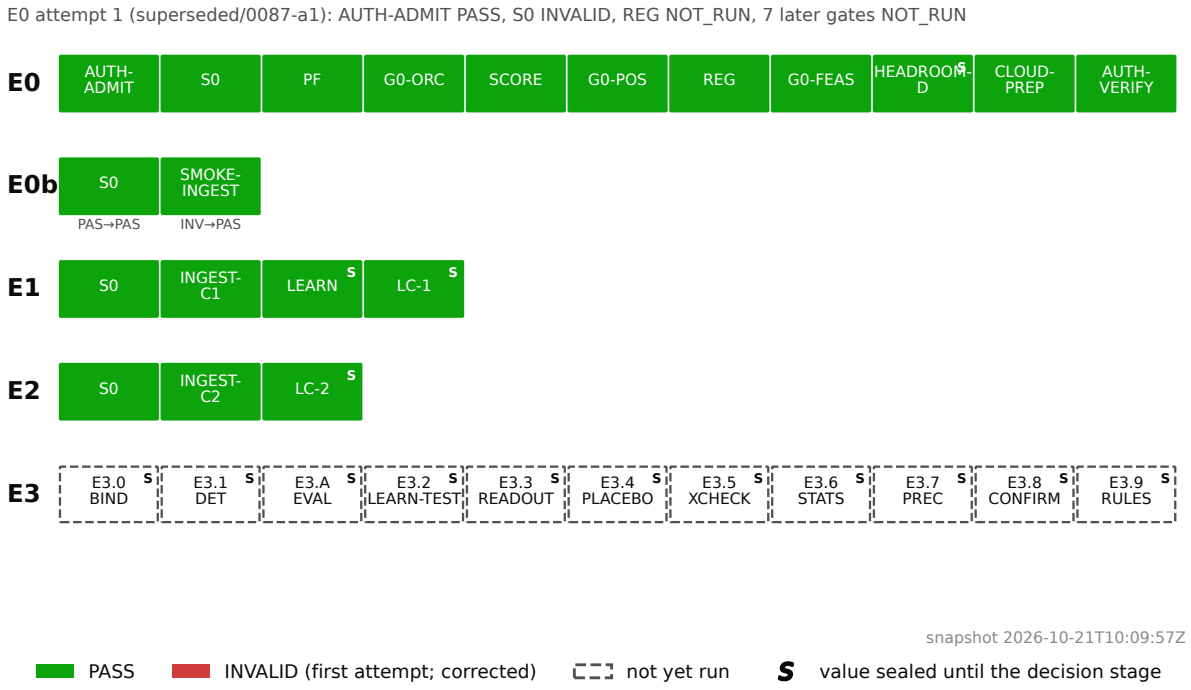


Figure 13: Crossing order in trained models, gate pipeline with statuses, under the protocol's gate names. Filled: PASS; outlined dashed: not yet run; "S": the gate's values are sealed until the decision stage; arrows below a box give the status history over attempts. Statuses and gate names only. Source: `x6-stage1-store/GATES.jsonl` and `superseded/0087-a1/GATES.jsonl` (fields gate, status, utc only; the values field is never read), via the snapshot `docs/research_summary/data/x6s1_status.json`; stage order from `X6S1-FROZEN-v1.md` §8.

addenda) and the coordinator's state (newest entry first). Codex acted as coordinator until 14 October, followed by a chain of Claude coordinator sessions, each handing over to a fresh one when its context grew too long. Each takeover wrote an assumption record, because release verifiers bind the issuer's identity; one takeover race was resolved by having the late session stand down. Workers were headless sessions or subagents launched per job, with context caps that forced rotation to fresh sessions; a transfer record registers each new worker identity. More than 100 jobs had been created by 26 October. Two operational hazards recurred: a shared subscription hitting usage limits (handled by waiting, never by switching accounts), and permission classifiers blocking some actions (never bypassed). I approved and installed an automatic-resume scheduler.

11.2 The cloud training pipeline and its integrity gates

The trained crossing-order study was the first trained on a GPU. The coordinator builds deterministic bundles (code archive, data, initial states, a ticket with seeds, steps and checkpoint interval, expected batch and RNG digests for every chunk) and releases a bundle hash to me, and I run a fixed notebook in the browser; only one cell (the bundle hash) may be edited. Every chunk records its batch digest and PCG64 state digest; the export carries a manifest whose hash I relay. Locally, four checks run per model before any evaluation: a weights check (manifest, inventory and canonical weights digests), a provenance check (every session and execution record matches the permitted cell templates, commit, protocol and bundle), a data-identity check (data, initial state, per-chunk batch and RNG digests), and a completeness check (a linked chain of exactly one record per checkpoint). An admission lock then fixes the model's only admissible chain. GPU training is not claimed to be reproducible; data, batch sequence, initial state and every evaluation are.

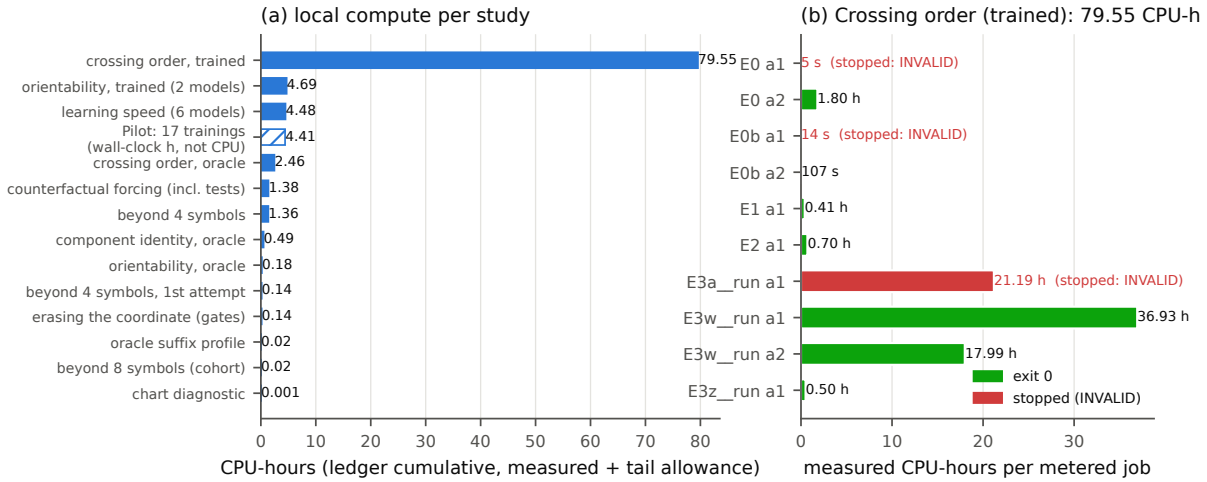


Figure 14: (a) Cumulative CPU-hours per study from each store's `cpu_ledger.json` (measured plus tail allowance; includes metered test suites); the hatched bar is the summed wall-clock time of the 17 pilot trainings, which predate the meter. (b) Trained crossing-order study, measured CPU per metered job and attempt, under the protocol's stage labels (E0 preparation; E0b smoke ingest; E1 and E2 ingest of the two training waves; E3a, E3w and E3z evaluation and decision; a1, a2 attempts). Sources: `<store>/cpu_ledger.json` for the stores named in `make_figures.py` (`LEDGERS`), snapshotted in `docs/research_summary/data/compute_ledgers.json`; `x6-stage1-store/cpu_ledger.json` (accounting fields only) via `data/x6s1_status.json`; `artifacts/MetricRecords/*/extra.json` (`wall_time_s`).

11.3 Defects found and corrected

The correction route was exercised many times, always before any outcome-bearing quantity existed or with documented independence from it. Examples: a 2π phase edge in the first four-symbol origin attempt; a module split under the 400-line rule in the orientability oracle study; an integer/string key mismatch and the ridge tolerance (discussed above) in counterfactual forcing; a `Fraction(dict)` call, a byte-order mark and a lazy sequence, test fixtures inside the project root and a missing reader in the four-symbol origin test; a key accumulation and an estimator sign error in the crossing-order oracle study; a constant path, a lineage check and a non-compiling pasted notebook cell in the trained crossing-order study; and, in the readout-attribution study, a lineage check that refused the stage's own entry records. Two classes of defect were caught only by real runs: those where tests exercised synthetic inputs but not the real output of a reused function (the `Fraction(dict)` case), and those that depend on the live environment (the pasted cell). Both led to added checks.

12 Compute

Figure 14 summarises local compute from the store ledgers. Including every metered run (preflight test suites among them), the follow-up studies used about 17.6 CPU-hours in the ledgers shown; the 17 pilot trainings took 4.4 wall-clock hours on a six-core laptop CPU (13–21 minutes each). The largest items are the two training studies (learning speed, six models; orientability, two models), the exact dynamic programme of the crossing-order oracle study, and the trained crossing-order study.

13 Current status and next steps

- **Running: readout attribution of the circle code** (§7). Frozen, authorised and implemented; its first stage is being rerun after a technical correction. No data from it exists yet.
- **Crossing order in trained models** is decided: INCONCLUSIVE (§10.2.1). Natural follow-ups for me to plan include the ZC-s1 designed-placebo response, the gap between the observed order sensitivity ($\tilde{\beta} \approx 0.06$) and the registered size (0.25) under longer training or larger models, and the missing retrain mechanism.
- **Other options** recorded by the coordinator and by me include an independent replication of the pilot's O models, a speed-matched successor of the learning-speed study, a redesign of the orientability learnability test, a follow-up on the four-symbol margin and comparator scale, and the remaining tasks of a seven-task project reorganisation. None is authorised.
- **Records out of date.** `state/PROGRESS.md` and the README's "Current follow-up status" date from 17 October and still describe the four-symbol origin test as running; it completed with EXCLUSION, the crossing-order oracle study completed with GO, and the trained crossing-order study has since been decided. This document reflects the coordinator's reviews, which are the more recent records.

14 Threats to validity and limitations

1. **No causal claim.** All pilot evidence is representational; both intervention designs stopped at gates.
2. **One toy setting.** One small architecture and recipe, synthetic HMMs on tiny lattices, a few seeds (three in the pilot, at most four per arm). Nothing is claimed about natural language or large models, other architectures, or populations of seeds.
3. **Statistics.** Pilot decisions use sign consistency and a data-derived envelope, not significance tests; panels are partitions within runs. Most later bounds are nominal bootstrap bounds; only the log-loss claims of the trained crossing-order study carry a finite-sample guarantee.
4. **Measurement choices.** Vietoris–Rips over $\mathbb{F}_2$, $M = 512$, the longest bar as S; four one-factor sensitivity variants. Circular coordinates use one cocycle and can give disconnected points arbitrary offsets.
5. **Exposure.** The circular audit, the counterfactual-forcing study, the four-symbol origin test and the readout-attribution study reuse the three pilot models, whose test panels had been unblinded; they draw fresh histories protected by registries, but the models are not fresh.
6. **Margins and controls.** The four-symbol margin sits above the oracle's own value; the P control is one-sided; the learning-speed kernel is confounded with speed.
7. **Reproducibility scope.** One machine and environment; one of 17 pilot trainings was retrained; the clean-rerun gate formally failed; GPU training in the crossing-order study is not reproducible by design.
8. **Process.** One post-data tolerance change (in the counterfactual-forcing study) occurred without my prior approval and I ratified it afterwards. Records written by agents were reviewed by other agents; I approved decision points but did not audit every computation.

15 Conclusion

Under controlled conditions, next-token training on a hidden walk produces a learned circle that reflects the world's topology, aligns with Bayesian belief geometry, and exceeds what the last symbol permits. Where it comes from is only partly known: the last-symbol embedding carries much of it, beliefs converge after a few symbols, and the beyond-four-symbol covariance of the learned phase is below a margin that even the oracle's phase does not reach. Whether the model uses it is unknown. The follow-up programme mostly produced well-defined stops rather than findings, and the stops taught something about one-sided controls, nonlinear encodings, non-specific interventions, fragile chart rules and margins set without oracle checks. The trained crossing-order study was designed to avoid each of these failure modes on a process where order provably matters. It reached its decision with every validity gate passed and found weak but reproducible natural order sensitivity in every seed, far below the registered effect size, so its preregistered verdict is INCONCLUSIVE and not a claim of order use. The readout-attribution study, now running, asks the use question again with an exact, linear attribution at the model's readout.

Table 1: Headline outcomes. "Label" is the outcome vocabulary of the governing frozen protocol. Sources: `README.md`; `reports/E2--E4`; `state/DECISIONS.md`; `agent-handoff/jobs/<job>/review.md`.

Study	Label	One-line reading
Pilot: a hole in the cylinder only (rule F1, smooth emission O)	SUPPORTED	$D > 0$ in 24/24 panels; median above envelope in 3/3 seeds
Pilot: geometry follows the full-history belief (H2)	sign-consistent	geometry tracks $b^{(32)}$ over $b^{(1)}$ in 48/48 panels
Pilot: the F1 rule under the permuted control P (H3)	REJECTED, uninformative	even the exact belief fails under P
Pilot: belief alignment under P (H3′)	SUPPORTED	ρ_s 0.06–0.11 $\rightarrow$ 0.70, beats mixing baselines
Pilot: the hole appears after history use (H4)	REJECTED	S rises at step 50, history use at step 100
Pilot: clean rerun from an empty store	FAIL	307/307 shared artifacts bitwise; one historical record missing
Two-step-ahead distinctions in the pilot world	INCONCLUSIVE / scoped NO-GO	no witnesses at the registered resolution
Erasing the circular coordinate (causal)	INCONCLUSIVE	linear eraser leaves nonlinear information; no matched controls
Geometry and learning speed	REJECTED	paired delays 250/0/0 steps (kernel confounded with speed)
Origin beyond the last 8, then 4, symbols (first attempts)	headroom FAIL / INCONCLUSIVE	no room at $K = 8$; chart gate failed at $K = 4$
Orientability needed only two steps ahead (oracle)	PASS	invisible one step ahead, needed two steps ahead
Component identity beyond short windows (oracle)	GO	component identity needs long, non-additive history
Orientability in trained models	NOT_LEARNED	both gate models failed the learnability criterion
Counterfactual state forcing (causal)	INCONCLUSIVE	intervention moved protected readouts; no matched controls
Origin beyond the last four symbols	EXCLUSION	$\lvert\Delta_4\rvert < 0.01$ in 3/3 models; oracle comparator 0.0085
Crossing order in the oracle	GO	order effect certified beyond every commutative statistic
Crossing order in trained models	INCONCLUSIVE	all validity gates PASS; mean $\tilde{\beta} = 0.061$, below the registered 0.25

Table 2: Guide to the studies. Each study is named in this document by the descriptive name in the first column. The internal label is the key used in the project records and in Appendix B; readers do not need it.

Study	Internal label	Question	Section
The pilot	E0–E4	Does training on the cylinder, and not the square, produce a learned loop?	5
Circular-baseline audit	—	Does the learned circle exceed what the last symbol permits?	6
Oracle suffix profile	—	How fast does the circular posterior converge with suffix length?	6
Origin beyond the last 8 symbols	K8	Is there room to test history beyond 8 symbols?	6
Origin beyond the last 4 symbols, first attempt	K4	Does the learned phase covary with history beyond 4 symbols?	6
Origin beyond the last 4 symbols	K4b	The same question with a repaired chart rule	6
Component identity beyond short windows (oracle)	X2a stage 0	Is long, non-additive history ever required?	6
Erasing the circular coordinate	X4-F	Does prediction depend on the circular coordinate?	7
Counterfactual state forcing	C1, C1′	Does the readout respond to a forced change of the phase?	7
Readout attribution of the circle code	C2	How much of the model's response to history runs through its circle code?	7
Two-step-ahead distinctions in the pilot world	X4 Phase B, B2	Do beliefs exist that agree one step ahead but not two?	8
Orientability needed only two steps ahead (oracle)	D-1	Can a topological distinction be invisible one step ahead?	8
Orientability in trained models	D-2a	Do trained models carry that distinction?	8
Geometry and learning speed	X5 G1	Does a curved kernel slow learning?	9
Crossing order in the oracle	X6 stage 0	Does a designed process carry order information?	10.1
Crossing order in trained models	X6 stage 1 (X6S1)	Do trained models use crossing order?	10.2

A Glossary

Term	Meaning
O, P, N, U	Emission conditions: smooth Gaussian; axis-permuted; uniform; one symbol per state
$b^{(32)}$, $b^{(1)}$	Exact belief after 32 symbols; posterior from the last symbol alone
F_1, F_2	Exact next-symbol law; joint law of the next two symbols
G_C, regret, captured	Context gain; mean KL from the oracle to the model; $1 - \text{regret}/G_C$
S, $D_{r,i}$, u_r	Normalised longest finite H_1 bar; double contrast; noise envelope
Δ_{geom}	Spearman alignment with $b^{(32)}$ minus alignment with $b^{(1)}$
F1, H2, H3, H3′, H4	The pilot's preregistered rule and hypotheses: hole in the cylinder only; geometry follows the full belief; F1 under P; belief alignment under P; the hole appears after history use
Baselines	Initialisation; last-symbol embedding; previous-token mixing; context average
m_K, B_K, r_4	Circular posterior mean from the last K symbols; $\mathbb{E}\|m_{32} - m_K\|$; $m_{32} - m_4$
Δ_4	Four-symbol origin estimand $\mathbb{E}[u \cdot r_4]$ for the learned chart phase u
ψ	Readout-attribution share: the part of the model's oracle-aligned response carried by the circle-code interchange
NA, AB, ZC	Crossing-order arms: the non-abelian process and its two commutative controls
Chart	Circular coordinate built from an H_1 cocycle on landmarks (LCC rule: largest component $\geq 95\%$)
Holonomy, local system	Composition of fibre permutations along a path; an assignment of permutations to edges, flat on a face if their product around it is the identity
Parikh vector	Counts of each directed seam crossing, ignoring order
$G_S(32)$	Excess log-loss at $t = 32$ of the best predictor that sees only statistic S
β_{ord}, C-ORD	Within-Parikh-class order slope; the primary order claim of the trained crossing-order study
SHARP, TARGET	Its log-loss claims: below $G_{\text{Parikh}}(32)$, and below 75% of it
Gate	A check that must pass before a stage's products are interpreted
Registry	Set of digests of every previously exposed history, used to certify fresh pools
Sealed	Written with a hash and released only at the decision stage or a stop
Technical correction	Coordinator-ruled fix of a documented defect; at most two jobs per defect
Release	Coordinator-signed permission for a specific stage of a study

B All stages, gates and outcomes

This table is the lookup key into the project records. It uses the internal study labels of Table 2, the protocols' own stage and gate names, and the numbers of the decision-log entries (D-. . .) and coordinator jobs whose reviews record each outcome.

Stage / study	Date	Decisive gate(s)	Outcome	Record
E0 (S1)	10-13	G-A	PASS	D-017; reports/E0
E1 (S2)	10-13	G-B, G-C, G-D	PASS; P one-sided	README (E1)
E2 (S3–S4)	10-13	G-E, G-F; F1 rule; H2	F1 SUPPORTED; H2 consistent	reports/E2
E3 (S5–S6)	10-13	F1 rule under P; H3′; H4	H3 rejected (uninf.); H3′ SUPPORTED; H4 REJECTED	reports/E3
E4 (S7–S8)	10-13	G-G; G-H	G-G PASS; G-H FAIL	D-049; reports/E4
X4 Phase B	10-13	coverage	INCONCLUSIVE	D-052
X4 Phase B2	10-14	witnesses	scoped NO-GO	D-054, D-055
Circular audit	10-14	descriptive	BEYOND_ SYMBOL_AND_ MIXING	D-057
X4-F	10-14	G3, G4	INCONCLUSIVE / FAMILY_ UNMATCHED	D-063
K8 cohort	10-14	oracle headroom	FAIL	x4-history-feas-store
X5 G1	10-14	H5 rule (dev)	REJECTED; closed (D-081)	job 0013; D-073
Oracle profile	10-14	exploratory	accepted, qualified	job 0014; D-074
K4	10-15	chart_calibration	INCONCLUSIVE	job 0018; D-078
C1	10-15	K4 chart dependency	blocked by freeze	job 0018 review
D-1	10-15	PF, S0, G0–G6	PASS	job 0020; D-080
Chart diagnostic	10-15	giant fraction ≥ 0.95	GO	job 0023
X2a stage 0	10-15	P3, P4, A3	GO_P1_ NONADDITIVE	job 0025; D-084
D-2a	10-15	learnability (stage 1)	NOT_LEARNED	job 0033
C1′ S3	10-16	G0, H1–H5, N0	PASS (after R11)	jobs 0039, 0041
C1′ S4	10-16	M1, M2-cal	INCONCLUSIVE / FAMILY_ UNMATCHED	jobs 0042, 0044
K4b E1	10-17	G0 … P1 (10 gates)	PASS (after R25, R26, R28)	jobs 0049–0055
K4b E2	10-17	H2-FRESH, T-PRE, ENDPOINT	EXCLUSION (3/3 models)	job 0057
X6 stage 0	10-18	PF, S0, G0–G7, D	GO_X6_NATURAL (after X7, X8)	jobs 0063–0066
X6S1 IMPL	10-20	gate 0	complete (freeze H)	job 0086
X6S1 E0	10-20	AUTH-ADMIT … CLOUD-PREP, AUTH-VERIFY	attempt 1 INVALID (S0); attempt 2 all PASS	jobs 0087, 0089
X6S1 E0b	10-20	SMOKE-INGEST	attempt 1 INVALID; attempt 2 PASS	jobs 0090, 0092
X6S1 E1	10-20	INGEST-C1, LEARN, LC-1	PASS; LEARN = GO	job 0093
X6S1 E2	10-20	INGEST-C2, LC-2	PASS; 12 models admitted	job 0094
X6S1 E3	—	E3.0–E3.9	all PASS; INCONCLUSIVE (rule 9)	0095–0099
C2 IMPL	10-22	gate 0; freeze	complete (FREEZE 24ff24cc)	jobs 0100–0106

Stage / study	Date	Decisive gate(s)	Outcome	Record
C2 S1	10-22, 10-26	release check; G0 (vii)	attempt 1 refused; attempts 2, 3 FAIL_ TECHNICAL (each corrected)	jobs 0107–0110

C Sources

- Pilot: `README.md`; `docs/RESEARCH_SUMMARY.md`; `docs/RESEARCH_PLAN_EN.md`; `PREREGISTRATION.md`; `PREREGISTRATION_X4.md`; `reports/E0` to `reports/E4`; `state/DECISIONS.md`.
- Follow-ups: `docs/proposals/`; the frozen protocols in `agent-handoff/planning/responses/`; the job reviews and reports in `agent-handoff/jobs/`.
- Coordination: `agent-handoff/COORDINATOR_STATE.md`; `agent-handoff/MORNING_REPORT_20261015_EN.md`; `agent-handoff/NEXT_RESEARCH_AND_WORKFLOW_EN.md`.
- Figure data and their provenance: `docs/research_summary/make_figures.py` and `docs/research_summary/data/`.

Paths starting with `agent-handoff/` or naming a `*-store` directory are execution records kept next to the repository, not in it. The review of each job is `agent-handoff/jobs/<job>/review.md`, with the job numbers of Appendix B.

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. arXiv:1610.01644, 2016.

[2] Jean-Yves Audibert, Rémi Munos, and Csaba Szepesvári. Exploration–exploitation tradeoff using variance estimates in multi-armed bandits. *Theoretical Computer Science*, 410(19): 1876–1902, 2009.

[3] Daniel Balcells, Andrew Jun Lee, Chirag Rastogi, Paul M. Riechers, Adam Shai, and Xavier Poncini. Large language models develop belief state geometry in-context. arXiv:2609.17376, 2026.

[4] David A. Barrington. Bounded-width polynomial-size branching programs recognize exactly those languages in NC^1. *Journal of Computer and System Sciences*, 38(1):150–164, 1989.

[5] Ulrich Bauer. Ripser: Efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5:391–423, 2021. arXiv:1908.02518.

[6] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

[7] Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. arXiv:2306.03819.

[8] Satwik Bhattamishra, Kabir Ahuja, and Navin Goyal. On the ability and limitations of transformers to recognize formal languages. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. arXiv:2009.11264.

[9] Alberto Bietti, Vivien Cabannes, Diane Bouchacourt, Herve Jegou, and Leon Bottou. Birth of a transformer: A memory viewpoint. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. arXiv:2306.00802.

[10] Omer Bobrowski and Primoz Skraba. On the universality of random persistence diagrams. arXiv:2207.03926, 2022.

[11] Cristian Bodnar, Francesco Di Giovanni, Benjamin Paul Chamberlain, Pietro Liò, and Michael M. Bronstein. Neural sheaf diffusion: A topological perspective on heterophily and oversmoothing in GNNs. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022. arXiv:2202.04579.

[12] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[13] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Naz Sepah, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Dmitriy Serdyuk, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, 2021. arXiv:2103.03098.

[14] Sasha Brenner, Thomas R. Knösche, and Nico Scherf. Predictive statistics shape emergent world representations of grid walkers. arXiv:2603.16689, 2026.

[15] Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2): 255–308, 2009.

[16] Bilal Chughtai, Lawrence Chan, and Neel Nanda. A toy model of universality: Reverse engineering how networks learn group operations. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023. arXiv:2302.03025.

[17] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.

[18] James P. Crutchfield and Karl Young. Inferring statistical complexity. *Physical Review Letters*, 63(2):105–108, 1989.

[19] Vin de Silva, Dmitriy Morozov, and Mikael Vejdemo-Johansson. Persistent cohomology and circular coordinates. *Discrete & Computational Geometry*, 45(4):737–759, 2011.

[20] Grégoire Delétang, Anian Ruoss, Jordi Grau-Moya, Tim Genewein, Li Kevin Wenliang, Elliot Catt, Chris Cundy, Marcus Hutter, Shane Legg, Joel Veness, and Pedro A. Ortega. Neural networks and the Chomsky hierarchy. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2207.02098.

[21] Benjamin L. Edelman, Ezra Edelman, Surbhi Goel, Eran Malach, and Nikolaos Tsilivis. The evolution of statistical induction heads: In-context learning Markov chains. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. arXiv:2402.11004.

[22] Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533, 2002.

[23] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

[24] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. Transformer Circuits Thread, `https://transformer-circuits.pub/2021/framework/index.html`, 2021.

[25] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. Transformer Circuits Thread; arXiv:2209.10652, 2022.

[26] Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2405.14860.

[27] Brittany Terese Fasy, Fabrizio Lecci, Alessandro Rinaldo, Larry Wasserman, Sivaraman Balakrishnan, and Aarti Singh. Confidence sets for persistence diagrams. *The Annals of Statistics*, 42(6):2301–2339, 2014.

[28] Stephen Fitz, Peter Romero, and Jiyan Jonas Schneider. Hidden holes: Topological aspects of language models. In *Proceedings of the NeurIPS Workshops on Symmetry and Geometry in Neural Representations*, volume 282 of *Proceedings of Machine Learning Research*, pages 948–980, 2026.

[29] Richard J. Gardner, Erik Hermansen, Marius Pachitariu, Yoram Burak, Nils A. Baas, Benjamin A. Dunn, May-Britt Moser, and Edvard I. Moser. Toroidal topology of population activity in grid cells. *Nature*, 602:123–128, 2022.

[30] Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What can transformers learn in-context? a case study of simple function classes. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022. arXiv:2208.01066.

[31] Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021. arXiv:2106.02997.

[32] Riccardo Grazzi, Julien Siems, Arber Zela, Jörg K. H. Franke, Frank Hutter, and Massimiliano Pontil. Unlocking state-tracking in linear RNNs through negative eigenvalues. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2411.12537.

[33] Wes Gurnee and Max Tegmark. Language models represent space and time. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.02207.

[34] Michael Hahn. Theoretical limitations of self-attention in neural sequence models. *Transactions of the Association for Computational Linguistics*, 8:156–171, 2020. arXiv:1906.06755.

[35] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019. arXiv:1808.01513.

[36] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.

[37] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. arXiv:2404.15255, 2024.

[38] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, 2018. arXiv:1709.06560.

[39] John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019. arXiv:1909.03368.

[40] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[41] Stuart H. Hurlbert. Pseudoreplication and the design of ecological field experiments. *Ecological Monographs*, 54(2):187–211, 1984.

[42] Herbert Jaeger. Observable operator models for discrete stochastic time series. *Neural Computation*, 12(6):1371–1398, 2000.

[43] Alexandru-Iulius Jerpelea. Planting a latent variable in natural-looking text: A more realistic test of belief states in LLMs and their link to concept geometry. arXiv:2608.26887, 2026.

[44] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. arXiv:2207.07048.

[45] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15:1–15:21, 2012.

[46] Kenneth Krohn and John Rhodes. Algebraic theory of machines. I. prime decomposition theorem for finite semigroups and machines. *Transactions of the American Mathematical Society*, 116:450–464, 1965.

[47] Belinda Z. Li, Zifan Carl Guo, and Jacob Andreas. (how) do language models track state? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2503.02854.

[48] Kenneth Li, Aspen K. Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.13382.

[49] Bingbin Liu, Jordan T. Ash, Surbhi Goel, Akshay Krishnamurthy, and Cyril Zhang. Transformers learn shortcuts to automata. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.10749.

[50] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1711.05101.

[51] Inbal Magar and Roy Schwartz. Data contamination: From memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. arXiv:2203.08242.

[52] Naman Malhotra, Jay Ambadkar, Abhinav Gupta, Kushal Kasivel, Abbas Schwarz, Kamillo Ferry, and Anthea Monod. Tracking representation dynamics in large language models with persistent homology. In *Proceedings of the 2nd Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2026)*, volume 334 of *Proceedings of Machine Learning Research*, pages 211–245, 2026.

[53] Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory (COLT)*, 2009. arXiv:0907.3740.

[54] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022. arXiv:2202.05262.

[55] William Merrill and Ashish Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023. arXiv:2207.00729.

[56] William Merrill, Jackson Petty, and Ashish Sabharwal. The illusion of state in state-space models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2404.08819.

[57] Marcus R. Munafò, Brian A. Nosek, Dorothy V. M. Bishop, Katherine S. Button, Christopher D. Chambers, Nathalie Percie du Sert, Uri Simonsohn, Eric-Jan Wagenmakers, Jennifer J. Ware, and John P. A. Ioannidis. A manifesto for reproducible science. *Nature Human Behaviour*, 1:0021, 2017.

[58] Gregory Naitzat, Andrey Zhitnikov, and Lek-Heng Lim. Topology of deep neural networks. *Journal of Machine Learning Research*, 21(184):1–40, 2020. arXiv:2004.06093.

[59] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2301.05217.

[60] Eshaan Nichani, Alex Damian, and Jason D. Lee. How transformers learn causal structure with gradient descent. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.14735.

[61] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.

[62] Joe O'Connor and Jacob Andreas. What context features can transformer language models use? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021. arXiv:2106.08367.

[63] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. Transformer Circuits Thread; arXiv:2209.11895, 2022.

[64] Rohit J. Parikh. On context-free languages. *Journal of the ACM*, 13(4):570–581, 1966.

[65] Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Yongyi Yang, Maya Okawa, Kento Nishi, Martin Wattenberg, and Hidenori Tanaka. ICLR: In-context learning of representations. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2501.00070.

[66] Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d'Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research*, 22(164):1–20, 2021. arXiv:2003.12206.

[67] Mateusz Piotrowski, Paul M. Riechers, Daniel Filan, and Adam S. Shai. Constrained belief updates explain geometric structures in transformer representations. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.01954.

[68] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. arXiv:2201.02177, 2022.

[69] Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

[70] Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. arXiv:2004.07667.

[71] Bastian Rieck, Matteo Togninalli, Christian Bock, Michael Moor, Max Horn, Thomas Gumbsch, and Karsten Borgwardt. Neural persistence: A complexity measure for deep neural networks using algebraic topology. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1812.09764.

[72] Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. arXiv:2310.18018, 2023.

[73] Yash Sarrof, Yana Veitsman, and Michael Hahn. The expressive capacity of state space models: A formal language perspective. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. arXiv:2405.17394.

[74] Adam Shai, Loren Amdahl-Culleton, Casper L. Christensen, Henry R. Bigelow, Fernando E. Rosas, Alexander B. Boyd, Eric A. Alt, Kyle J. Ray, and Paul M. Riechers. Transformers learn factored representations. arXiv:2602.02385, 2026.

[75] Adam S. Shai, Sarah E. Marzen, Lucas Teixeira, Alexander Gietelink Oldenziel, and Paul M. Riechers. Transformers represent belief state geometry in their residual stream. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. arXiv:2405.15943.

[76] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4):817–879, 2001.

[77] Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011.

[78] Satinder Singh, Michael R. James, and Matthew R. Rudary. Predictive state representations: A new theory for modeling dynamical systems. In *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2004. arXiv:1207.4167.

[79] Koustuv Sinha, Robin Jia, Dieuwke Hupkes, Joelle Pineau, Adina Williams, and Douwe Kiela. Masked language modeling and the distributional hypothesis: Order word matters pre-training for little. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021. arXiv:2104.06644.

[80] Keyon Vafa, Justin Y. Chen, Ashesh Rambachan, Jon Kleinberg, and Sendhil Mullainathan. Evaluating the world model implicit in a generative model. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. arXiv:2406.03689.

[81] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. arXiv:1706.03762.

[82] Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2111.02080.

[83] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020. arXiv:2002.04745.

[84] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.16042.

[85] Zhiyu Zhang and Yupeng Li. Tracking states or tracking cosets? an algebraic account of learned state tracking. arXiv:2609.29951, 2026.

[86] Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. The clock and the pizza: Two stories in mechanistic explanation of neural networks. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. arXiv:2306.17844.

[87] Donglin Zhuang, Xingyao Zhang, Shuaiwen Leon Song, and Sara Hooker. Randomness in neural network training: Characterizing the impact of tooling. arXiv:2106.11872, 2021.

[88] Afra Zomorodian and Gunnar Carlsson. Computing persistent homology. *Discrete & Computational Geometry*, 33(2):249–274, 2005.